Subject-Dominance Limits (SDL) in Clinical Resting-State EEG Foundation Models

Draft compilation — 2026-04-20 — Path A (clinical/engineering framing)

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Source data and scripts

- Master FT-vs-LP table: `paper/figures/source_tables/master_frozen_ft_table_v2.md`
- FOOOF ablation results: `results/studies/fooof_ablation/SUMMARY.md`
- Per-window LP results: `results/studies/perwindow_lp_all/SUMMARY.md`
- FT vs LP paired comparison: `results/studies/ft_vs_lp/comparison.json`
- Strategy: `docs/paper_strategy_v3_minimum_viable.md`

Figures embedded

- Fig 4.1 Variance atlas (§4)
- Fig 5.1 Honest-evaluation funnel (§5); Fig 5.2 Permutation-null density (§5.3)
- Fig 6.1 Paired contrast (§6); Fig 6.2 FOOOF ablation (§6.5)
- Fig 7.1 Architecture ceiling (§7); Fig 7.2 Classical baselines (§7.2)

SDL Paper Draft — Abstract + §1 Introduction (v2, 2026-04-20)

Framing: Path A — clinical-engineering audience (~60% clinical / ~40% engineering). Target venue: *Journal of Neural Engineering*, *Brain Informatics*, *IEEE TNSRE*, or *Frontiers in Neuroinformatics*. Tone: practical, decision-oriented, concrete. Avoids pure-ML framing; leads with the clinical practitioner's question.

Supersedes: the original thesis-format abstract and §1 in `sdl_paper_draft_sections_1-3.md` (2026-04-15).

Abstract

Electroencephalography foundation models (EEG-FMs) have reported strong performance on diagnostic benchmarks, rekindling interest in applying them to small clinical cohorts. We ask a practical question facing any clinical researcher with a 20–500-subject resting-state EEG dataset: *can I fine-tune an EEG-FM on my data and expect a usable model?* Across three publicly available pretrained FMs (LaBraM, CBraMod, REVE) and four clinical cohorts (UCSD Stress, ADFTD Alzheimer's, TDBRAIN MDD, EEGMAT mental arithmetic), evaluated under matched per-window linear probing and per-FM canonical fine-tuning with subject-disjoint cross-validation, we find that fine-tuning exceeds a frozen linear probe in only 1 of 12 cells (LaBraM × ADFTD, +5.6 pp); 2 of 12 show statistically suspect apparent rescues on a small imbalanced cohort (Stress, likely attributable to label–subject collinearity); and 6 of 12 cells show fine-tuning *degrading* performance below frozen linear probing. A published benchmark on UCSD Stress reporting 0.90 balanced accuracy drops to 0.43–0.55 under honest subject-level evaluation. Using controlled aperiodic-vs-periodic ablation based on FOOOF spectral parameterisation, we demonstrate causally that it is the 1/f aperiodic background — not periodic oscillations — that carries subject identity in frozen FM embeddings: removing the aperiodic component collapses subject-identity decodability (up to −26 percentage points, REVE on EEGMAT), whereas removing periodic peaks leaves it unchanged (≤ 0.7 pp). We distil these findings into a three-step pre-flight checklist for clinical EEG-FM deployment: (1) verify the target label has a published within-recording neural anchor; (2) measure frozen-LP balanced accuracy under subject-disjoint cross-validation; (3) check subject-identity decodability against state-label decodability to flag shortcut risk. Clinical researchers with cohorts lacking a published neural anchor should expect classification ceilings near 0.55 BA regardless of FM choice or fine-tuning effort; classical band-power baselines are within 5 pp of anything our three FMs achieved on that label class.

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§1 Introduction

Resting-state electroencephalography (rsEEG) is being increasingly proposed as a low-cost, accessible neurophysiological biomarker for psychiatric and neurological conditions, from Alzheimer's disease to depression to chronic stress. The clinical appeal is obvious: a 5–10 minute eyes-closed recording on a 19-channel cap costs a fraction of MRI and can be deployed in primary care or remote settings. The scientific question — whether rsEEG contains enough individual-level signal to distinguish clinical states across subjects — is less settled.

Recent work on large EEG foundation models (FMs) has argued the answer is yes. LaBraM [Jiang et al. 2024], CBraMod [Wang et al. 2024], REVE [Zhong et al. 2025], and related architectures train on thousands of hours of aggregated public EEG data and report strong performance on downstream clinical benchmarks. In one representative example, Wang et al. 2025 (arXiv:2505.23042) report 90.47 % balanced accuracy for fine-tuned LaBraM on a 17-subject rsEEG stress cohort [Komarov, Ko, Jung 2020]. Taken at face value, this result would imply that modern EEG FMs approximate a useful clinical classifier for an affective state on a cohort of fewer than 100 recordings — a regime in which traditional signal processing has historically struggled.

A clinical researcher reading such a result faces a concrete decision: should I buy a GPU, fine-tune this FM on my 50-subject Parkinson's cohort, and expect the reported performance levels to transfer? This paper answers that question empirically, and the answer is: **not without verification that is rarely reported in the benchmark literature**.

1.1 What we find

Across 12 (FM × dataset) cells — three pretrained EEG FMs evaluated on four publicly available clinical rsEEG cohorts (UCSD Stress 17 subjects, ADFTD 65 subjects, TDBRAIN 359 subjects, EEGMAT 36 subjects) — under subject-disjoint cross-validation and matched per-window evaluation protocols for frozen linear probing versus fine-tuning, we find four consistent patterns:

1. **Fine-tuning rarely exceeds a frozen linear probe.** Only 1 of 12 cells shows fine-tuning providing a clinically meaningful rescue (+5.6 pp on LaBraM × ADFTD). Three cells are tied within 1 pp. Six cells show fine-tuning *degrading* classification accuracy below the frozen baseline.

2. **Single-dataset published benchmarks can collapse by 20–40 pp under honest evaluation.** On the UCSD Stress cohort [Komarov 2020], the previously reported 0.90 BA falls to 0.43–0.53 across three FMs when the evaluation protocol is replaced by subject-disjoint cross-validation, and to null-indistinguishable under permutation-label control (LaBraM FT $p = 0.70$). No architecture we tested — from a 3-thousand-parameter 2018 compact CNN to a 1.4-billion-parameter 2025 foundation model — exceeds this honest ceiling.

3. **Apparent "FT rescues" on small imbalanced cohorts are mechanistically suspect.** Two of the three positive ΔFT values in our grid occur on the 17-subject Stress cohort with 14 positives concentrated in a subset of subjects. This structural collinearity between subject identity and recording label offers FT a shortcut: predict label by matching subject-fingerprint. Verifying whether these rescues are real signal acquisition or shortcut exploitation requires a permutation-null control that is rarely reported.

4. **Subject identity in frozen FM embeddings is causally carried by aperiodic 1/f structure.** Using a FOOOF-based ablation that surgically removes the aperiodic background from EEG input and re-extracts FM features, we show that subject-identity decodability from the frozen embedding drops by up to 26 percentage points when aperiodic is removed but remains essentially unchanged when periodic oscillatory peaks are removed. The 1/f background — a physiological trait property shaped by cortical E/I balance and volume conduction [Donoghue et al. 2020 Nat Neurosci; Gao et al. 2017] — is the substrate of subject dominance in modern EEG FM representations. Periodic oscillations are not.

1.2 What this means for clinical deployment

These four patterns have a single joint implication for clinical EEG-FM use: **whether a given (FM $\times$ cohort $\times$ label) combination will produce a useful clinical classifier is determined mostly by properties of the target label and cohort, not by the FM or by the fine-tuning effort**. We operationalise this in a three-step pre-flight checklist (§9):

- **Check 1** (literature search, < 1 hour): Does the target label have a published within-recording neural anchor — a peer-reviewed epoch-level EEG effect tied to the label category?
- **Check 2** (frozen LP, < 2 hours compute): Under subject-disjoint CV, does the frozen FM embedding linearly separate the target label at BA > 0.55?
- **Check 3** (subject-identity probe, minutes): Does the FM embedding also linearly separate subject identity at rates that dominate state-label separability? If so, any fine-tuning gain is shortcut-suspect until a permutation-null control confirms otherwise.

On the four cohorts studied, only EEGMAT (rest vs arithmetic, alpha-ERD anchor) and ADFTD (Alzheimer's, alpha-slowing anchor) pass all three checks. TDBRAIN and Stress fail Check 1, and reported FM benchmarks on these cohorts should be interpreted with suspicion.

1.3 Scope and contribution

This paper does not propose a new architecture, a new pretraining objective, or a new dataset. We propose three things:

1. **A diagnostic protocol** for auditing published EEG-FM benchmarks on small clinical cohorts (§5–§8). Applied to the UCSD Stress benchmark, the diagnostic accounts for a 20–40 pp accuracy inflation that previously lacked an explanation.
2. **A causal demonstration** — via FOOOF aperiodic/periodic ablation (§6.5) — that subject dominance in EEG FM embeddings is carried by the 1/f background and not by periodic oscillations. This is, to our knowledge, the first causal evidence on FM embeddings for a claim previously supported only by correlational EEG-biometric studies [Demuru & Fraschini 2020; Lanzone et al. 2023].
3. **A deployment checklist** for clinical researchers and reviewers (§9) that turns §5–§8's findings into three concrete pre-flight checks. The checklist is cheap to run (< 1 day of compute on a consumer GPU for typical cohorts) and can be reported as a standard supplementary diagnostic alongside any EEG-FM clinical benchmark.

1.4 Organisation

§2 situates this work relative to the EEG-FM benchmark literature, subject-leakage warnings (BENDR, AdaBrain-Bench, Brookshire 2024), and aperiodic-as-fingerprint work (Demuru 2020, Donoghue 2020). §3 describes datasets, foundation models, and protocols. §4 establishes the frozen subject-atlas structure of the three FMs across four datasets. §5 presents the honest re-evaluation of the UCSD Stress benchmark. §6 reports the paired EEGMAT–Stress comparison, cross-architecture band consensus, and FOOOF-based causal ablation. §7 establishes architecture-independence of the Stress ceiling across seven models spanning six orders of magnitude in parameter count. §8 presents the cross-dataset FT-versus-LP grid. §9 presents the clinical pre-flight checklist. §10 lists limitations, open questions, and future work.

SDL Paper Draft — §2 Related Work (v2, 2026-04-20)

Revisions from v1: (a) 20+ verified citation additions from 2026-04-19 literature survey (`docs/paper_strategy_v3_minimum_viable.md` §9 anchor list); (b) new subsection on aperiodic-as-fingerprint lineage (Demuru 2020, Lanzone 2023, Gao 2017, Donoghue 2020) supporting §6.5 FOOOF causal ablation; (c) updated EEG-FM critique lineage with Yang ICLR 2026, EEG-FM-Bench, arXiv:2601.17883, Brain4FMs, AdaBrain-Bench. All citations URL-verified on 2026-04-19.

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§2 Related Work

2.1 EEG foundation models and their clinical evaluation

The past two years have seen emergence of large EEG foundation models pretrained on aggregated public corpora. **LaBraM** [Jiang et al. 2024, ICLR; arXiv:2405.18765] uses a vector-quantised neural tokenizer with a 12-layer transformer, pretrained on ∼2,500 hours of EEG across 20 datasets (5.8M to 369M parameters). **CBraMod** [Wang et al. 2024/2025, ICLR 2025; arXiv:2412.07236] applies criss-cross attention for channel × time modelling. **REVE** [Zhong et al. 2025] scales to ∼1.4 billion parameters with linear patch embedding. **EEGPT** [Wang et al., NeurIPS 2024; OpenReview `lvS2b8CjG5`] introduces masked reconstruction with 10M parameters and chooses linear probing as its headline metric. **BIOT** [Yang et al., NeurIPS 2023; arXiv:2305.10351], **Brant** [NeurIPS 2023], and **NeuroLM** [Jiang & Wang, ICLR 2025; arXiv:2409.00101] span the rest of the landscape. Each reports competitive or state-of-the-art results on selected downstream tasks.

LEAD [Wang et al. 2025/2026, arXiv:2502.01678] is the dedicated Alzheimer's EEG foundation model most directly relevant to our ADFTD and Stress evaluations. LEAD reports LaBraM 91.1 % subject F1 on "ADFTD" under 8:1:1 subject-independent split with 5 seeds. The ADFTD cohort used by LEAD is a merged corpus of ds004504 (resting-state) and ds006036 (photic-stimulation); the RS-only number reported in LEAD's own Table 3 is 55.49 ± 15.08 %. Our ADFTD evaluation uses ds004504 (RS-only, the clinically deployable paradigm), which places our numbers in the comparable range to LEAD's RS-only benchmark, not its merged-paradigm headline.

2.2 Published critiques and benchmarks of EEG foundation models

Recent benchmarks have converged on a consistent finding: EEG FMs do not uniformly outperform simpler baselines under honest subject-disjoint evaluation.

"Are EEG Foundation Models Worth It?" [Yang, Sun, Li, Van Hulle, ICLR 2026; OpenReview `5Xwm8e6vbh`] benchmarks 7 EEG FMs across 6 evaluation protocols and reports that FT-FMs do not consistently beat compact models or classical decoders in data-scarce settings, linear probing is "consistently weak", and no clear scaling law holds.

EEG-FM-Bench [Xu et al. 2025/2026; arXiv:2508.17742] evaluates 7 FMs plus 2 general time-series models across 14 datasets × 10 paradigms and confirms gradient conflict between masked reconstruction and downstream classification objectives, as well as scaling

laws that deviate from typical transformer trends (compact architectures with task-relevant inductive biases often outperform larger FMs).

EEG Foundation Models: Progresses, Benchmarking, Open Problems [arXiv:2601.17883, Jan 2026] surveys 12 FMs × 13 datasets × 9 paradigms under leave-one-subject-out (LOSO) and few-shot conditions, concluding that larger FMs do not consistently outperform specialist scratch models and that linear probing is often insufficient.

Brain4FMs [arXiv:2602.11558, Feb 2026] benchmarks 15 brain foundation models across 18 datasets and reports affect and communication tasks failing cross-subject, with supervised baselines exceeding most BFMs on emotion recognition.

AdaBrain-Bench [arXiv:2507.09882 v2 Aug 2025] reports EEGPT under Cross-Subject Transfer on BCI-IV-2a: LP 47.89 % versus FT 25.81 % — a 22 pp degradation under full fine-tuning, framed as large-model overfitting on cross-subject distribution shift.

EEG Foundation Models: Critical Review [arXiv:2507.11783 v3 Jul 2025 / NeurIPS 2025] presents the most explicit position-paper critique, noting that masked-reconstruction pretraining may encourage noise memorization and that contrastive, cross-modal, and event-aware objectives are more robust.

This prior critique lineage establishes that the FT-underperforms-LP pattern (and the broader weaker-than-claimed cross-subject performance of EEG FMs) is a published observation. Our contribution is orthogonal: we provide a **mechanistic explanation** for why the pattern occurs (§6.5 aperiodic-carried subject dominance) and a **deployment-oriented diagnostic** (§9 pre-flight checklist) for clinical researchers contemplating EEG-FM use on their own cohorts.

2.3 Subject leakage and honest cross-validation in EEG deep learning

The impact of subject leakage on EEG-ML accuracy inflation has been documented for nearly a decade. **Kostas et al. 2021** [BENDR, Frontiers in Human Neuroscience; arXiv:2101.12037] established leave-one-subject-out and leave-multi-subjects-out as standard protocols in EEG SSL; their Fig 3 / Table 2 report linear classification *outperforming* full fine-tuning in 4 of 5 datasets under LOSO — the original FT < LP observation that our §8 replicates at scale.

Banville, Chehab, Hyvärinen, Engemann, Gramfort 2021 [J. Neural Eng. 18(4) 046020] formalise subject-stratified evaluation in clinical EEG SSL. **Gemein et al. 2020** [NeuroImage] benchmark TUH abnormal detection under subject-disjoint splits, establishing the reference baseline. Most directly, **Brookshire et al. 2024** [Frontiers in Neuroscience, DOI 10.3389/fnins.2024.1373515] survey 63 post-2018 translational EEG DL studies (AD, PD, ADHD, depression, schizophrenia, seizure) and show that segment-level holdout systematically inflates accuracy by 15–30 pp versus subject-level holdout. Brookshire 2024 is the canonical current reference for segment-vs-subject CV inflation in clinical EEG deep learning.

Our Wang et al. 2025 audit (§5) fits directly into this lineage: a single protocol correction (subject-level `StratifiedGroupKFold(5)` replacing trial-level `StratifiedKFold(5)`) accounts for ∼20 pp of the reported 40 pp gap. The residual gap we leave explicitly as unresolved, attributed to unspecified Wang pipeline design choices we cannot audit without full training logs.

2.4 Variance decomposition and representation diagnostics

Gibson, Lobaugh, Joordens, McIntosh 2022 [NeuroImage 252; pii/S105381192200163X]

partition EEG signal variance into across-subject and across-block components at the raw-signal level, finding across-subject variance dwarfs across-block (task) variance. This is the nearest published precedent to our §4 η^2 nested variance decomposition, with a key distinction: Gibson 2022 operates on raw EEG features, while our §4 operates on frozen FM embeddings. Quantifying subject-vs-label variance structure *inside* pretrained FM representations using η^2 nested SS decomposition is, to our knowledge, previously unclaimed.

Wagh, Varatharajah et al. 2022 [NeurIPS; dl.acm.org/doi/10.5555/3600270.3601807] report the closest-in-spirit diagnostic-framework paper for EEG-ML latent spaces under distribution shift; our §4 subject atlas applies analogous analysis specifically to EEG-FM representations on clinical rsEEG.

EEG-FM-Bench [Xu et al. 2025, arXiv:2508.17742] uses CKA, RSA, and subspace affinity between FM layers during FT to track representation evolution, reporting cross-subject balanced-accuracy drops of 15, 19, and 7 pp for LaBraM, CBraMod, and EEGPT respectively on SEED. Our §8 complements this finding at the representation-level (CKA / RSA against task labels and layer-to-layer, as Xu et al. do) with *balanced accuracy* at the downstream classification level on rsEEG clinical tasks.

GC-VASE [Mishra et al., arXiv:2501.16626] reports subject-identification decoding as a *desired* VAE property (89.81 % bAcc on ERP-Core, 70.85 % on SleepEDFx-20), embracing subject identity as signal rather than confound. This is an illuminating inversion of our SDL framing: the exact representation property that GC-VASE architects for is what §4 shows EEG FMs already have as an emergent property, with downstream cost to clinical discrimination.

2.5 Aperiodic 1/f spectral structure and EEG biometric fingerprinting

The physiological grounding for aperiodic 1/f structure comes from **Donoghue et al. 2020** [Nat Neurosci 24:1655-1665; nature.com/articles/s41593-020-00744-x], which introduces FOOOF as a method for parameterising neural power spectra into periodic (Gaussian peaks) and aperiodic (1/f) components. **Gao, Peterson, Voytek 2017** [NeuroImage; PMID 28676297] link the aperiodic exponent to cortical excitation-inhibition balance. **Waschke, Kloosterman, Obleser, Garrett 2021** [Neuron 109(5):751-766; cell.com/neuron] review the link between aperiodic variability and behavior.

The aperiodic-as-subject-fingerprint claim originates in **Demuru & Fraschini 2020** [arXiv:2001.09424 / PMID 32421651] for EEG and was replicated on MEG by **Lanzone et al. 2023** [NeuroImage; pii/S1053811923004111]. Both papers show that 1/f slope and offset alone identify subjects with near-ceiling accuracy and that the aperiodic component dominates subject-identifiability over periodic peaks. **Voytek et al. 2024** [Nat Commun; nature.com/articles/s41467-024-45922-8] provide biophysical grounding for why aperiodic EEG carries individual information.

The classical EEG biometric literature [Campisi & La Rocca 2014; Maiorana, Marcel, Ruiz-Blondet et al. 2015-2025] established that subject recognition from resting EEG approaches near-ceiling accuracy with classical features. What is new in our §6.5 is a causal demonstration on *pretrained FM embeddings* rather than correlational evidence on classical features, and the dissociation between aperiodic (subject-carrying) and periodic (subject-neutral) spectral content.

2.6 Subject-invariant representation learning in EEG

Methods attempting to remove subject identity from EEG representations — typically as a

confound to enable better cross-subject task performance — include: adversarial domain-invariant training [**Özdenizci, Wang, Koike-Akino, Erdoğmuş 2020**, IEEE Access; PMC7971154], VAE + domain-adversarial topographic embedding [**Han, Etemad et al. 2021**, IEEE J-BHI; PMC7961341], mutual-information-based MLP subject disentanglement [**Zhang & Liu 2025**, arXiv:2501.08693], and per-subject low-rank adaptation [arXiv:2510.08059, 2025]. These methods treat subject identity as a nuisance variable to be removed. GC-VASE (§2.4) treats it as signal to be preserved.

Our framing is diagnostic rather than corrective: we quantify subject dominance without removing it, and argue that the *quantification itself* is a prerequisite for clinical deployment decisions. The pre-flight checklist in §9 places no prior preference on preserving or removing subject identity — it simply makes the trade-off visible to the clinical researcher.

2.7 Within-subject neural contrasts and task anchors

The neurophysiological basis for distinguishing EEGMAT (rest vs mental arithmetic) from DASS-based stress labels as targets for within-subject FM evaluation rests on decades of oscillatory EEG literature. **Klimesch 1999** [Brain Research Reviews] established alpha desynchronisation as the canonical cognitive-load signature. **Pfurtscheller & Aranibar 1979** provided the event-related desynchronisation framework. **Klimesch 2012** [NeuroImage] updated the oscillatory model. For Alzheimer's EEG, **Babiloni et al. 2020** [Clin Neurophysiol] and **Ieracitano et al. 2021** document alpha slowing as a robust within-recording signature.

By contrast, no peer-reviewed EEG literature establishes a per-recording within-subject neural correlate of DASS-21 or Daily Stress Scale at the temporal resolution used in per-recording classification benchmarks. This absence — rather than the presence of a published anchor — is what our §6 paired experiment operationalises and what §9 Check 1 codifies.

2.8 Small-sample reliability in EEG deep learning

A multisite study of N = 2,874 rsEEG recordings [bioRxiv 2025.11.10.687610] concludes that "results from small samples were unstable, with inflated and highly variable effect sizes across iterations. Larger samples produced consistent findings, converging on uniformly small but robust effects." An independent brain-behavior correlation study [bioRxiv 2026.02.06.704323] reaches the same conclusion from a different angle. Our UCSD Stress cohort (70 recordings, 14 positive, 17 subjects) sits at the fragile end of this spectrum; the §5.3 permutation-null result ($p = 0.70$) quantifies the fragility.

2.9 Positioning

Against this literature, this paper's contribution is:

- **Not** a new EEG-FM architecture or pretraining objective;
- **Not** another independent benchmark of existing FMs;
- **Rather**: a diagnostic protocol (§5–§6.5), a causal mechanism (§6.5 FOOOF ablation), and a clinical deployment checklist (§9) that integrate existing observations into a directly usable tool for a clinical researcher or benchmark reviewer.

The closest precedents are Brookshire et al. 2024 (segment-leakage warning, extended to FMs here) and Yang et al. ICLR 2026 ("Are EEG FMs Worth It?" — which answers "often not"; we extend the "why" to a specific mechanism and give a "when"-to-skip-them rule).

SDL Paper Draft — §3 Methods (v2, 2026-04-20)

Revisions from v1: adds (a) per-window matched-protocol LP description (§3.5); (b) canonical per-FM FT recipe description (§3.6); (c) FOOOF-based signal ablation pipeline (§3.7); (d) LOSO robustness check (§3.8); (e) permutation-null control procedure (§3.9). §3.1-§3.4 from v1 are lightly updated for consistency.

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§3 Methods

3.1 Datasets

We evaluate on four publicly available clinical rsEEG cohorts:

UCSD Stress [Komarov, Ko, Jung 2020, IEEE TNSRE 28(4):795]. 17 Taiwanese graduate students, 92 recordings with per-recording DASS-21 and Daily Stress Scale self-reports. We use the 70 recordings with both DASS-21 and DSS present (`data/comprehensive_labels.csv`). Per-recording binary label via DASS-21 threshold > 50 ("Group" column: 14 positive / 56 negative), matching the protocol in Wang et al. 2025. 5-second epochs at 200 Hz $\times$ 30 channels from the original Curry 8 cap.

ADFTD [Miltiadous et al. 2024, OpenNeuro ds004504, RS-only]. 65 subjects, 195 recordings. AD + FTD vs healthy controls (binary positive/negative: 108/87). 19 channels resampled to 200 Hz, 5-second epochs. Used here in the clinically deployable RS-only configuration (not the merged RS+PS corpus used by LEAD).

TDBRAIN [van Dijk et al. 2022]. 359 subjects, 734 recordings. MDD vs healthy controls (`target_dx="MDD"`, `condition="both"` combining EO and EC). Binary 640 positive / 94 negative. 26 channels downsampled to 19 for cross-FM compatibility (COMMON_19 subset), 200 Hz, 5-second epochs.

EEGMAT [Zyma et al. 2019, PhysioNet]. 36 healthy subjects $\times$ 2 recordings (resting baseline + serial-subtraction mental arithmetic; 72 recordings, 36 paired). 19 channels, 200 Hz, 5-second epochs.

All four datasets are accessed via `pipeline/*_dataset.py` adaptors that normalise to (M, C, T) epoch tensors with per-FM input norm (§3.4) applied.

3.2 Foundation models

Three pretrained EEG FMs:

- **LaBraM** [Jiang et al. 2024 ICLR; braindecode/labram-pretrained HuggingFace weights]: 5.8M-parameter (Base), 12-layer transformer with vector-quantised neural tokenizer and 128-slot channel position embedding. Output: (B, 200).
- **CBraMod** [Wang et al. 2024/2025 ICLR 2025; open weights `cbramod-base`]: criss-cross attention transformer. Output: (B, 200).
- **REVE** [Zhong et al. 2025; open weights `reve-base`]: ~1.4B-parameter transformer with linear patch embedding. Output: (B, 512).

All three are used in frozen mode and fine-tuning mode via a shared `baseline/abstract/factory.py` interface. Channel mapping for 19-channel datasets uses COMMON_19 subset (`pipeline/common_channels.py`).

3.3 Baseline architectures for cross-architecture robustness (§7)

For Stress cohort comparison: EEGNet [Lawhern 2018, ~3k params]; ShallowConvNet [Schirrmeister 2017, ~40k]; DeepConvNet [Schirrmeister 2017, ~300k]; EEGConformer [Song 2023, ~800k]. For classical band-power baselines (§5.2): Random Forest, LogReg-L2, SVM-RBF, XGBoost, all with class weight balancing (`class_weight='balanced'` for RF/LogReg/SVM, `sample_weight=compute_sample_weight('balanced', y_train)` for XGBoost).

3.4 Per-FM input normalisation

Input normalisation is **per-FM, not global**, following each FM's original pretraining convention (documented in `CLAUDE.md §2`):

- **LaBraM**: `norm='zscore'` per channel, per window. Matches the paper's fine-tuning recipe.
- **CBraMod**: `norm='none'`. The extractor internally applies `x / 100` in `cbramod_extractor.py:102`; pre-zscoring would double-scale input to ~0.
- **REVE**: `norm='none'`. The patch embedding is a `Linear(patch_size → embed_dim)` trained on µV-scale pretraining data.

A global `norm` flag across multiple FMs is **never** used; getting this wrong silently destroys the run. Cache directories encode norm: `data/cache` (zscore) and `data/cache_nnone` (no norm).

3.5 Matched per-window linear probing protocol

The frozen linear probe (LP) protocol used throughout §5–§8 is designed to be *apples-to-apples with fine-tuning*: both train on per-window samples, both pool predictions per recording for recording-level metrics. The alternative protocol — averaging window-level features per recording, then training a classifier on recording-level samples — artificially inflates LP accuracy relative to FT because it discards within-recording variance that FT uses during training. We avoid this protocol throughout.

Concretely, for each (FM, dataset) cell:

1. **Feature extraction (per window).** Load the dataset in its native configuration (channels, norm per §3.4, 5-s epochs). For each epoch, run the frozen FM to produce a per-window embedding of dimension 200 (LaBraM, CBraMod) or 512 (REVE). Save (features, window_rec_idx, window_labels, window_pids) to `results/features_cache/frozen_{model}_{dataset}_perwindow.npz`.

2. **Subject-level CV at recording level.** Use `StratifiedGroupKFold(n_splits=5, shuffle=True, random_state=seed)` with `groups=patient_id` to partition **recordings** into 5 train-test folds stratified by recording label.

3. **Per-window training within each fold.** The training fold's recordings contribute all their windows as training samples; the test fold's recordings contribute all their windows as test samples. Feature preprocessing: per-dim 1-99 percentile clip (fit on train fold only), then `StandardScaler` (fit on train fold only). Classifier:

`LogisticRegression(C=1.0, class_weight='balanced', solver='liblinear', max_iter=5000, tol=1e-3)`.

4. **Prediction pooling.** For each test-fold recording, average the per-window class-1 probabilities across all its windows → pooled probability. Threshold at 0.5 → recording-level prediction. This matches `train_ft.py`'s global prediction pooling.

5. **Recording-level balanced accuracy.** Compute `balanced_accuracy_score(rec_labels, rec_pred)` pooled across all 5 folds.

6. **Multi-seed aggregation.** Repeat (2)–(5) for 8 seeds: 42, 123, 2024, 7, 0, 1, 99, 31337. Report mean and sample std (ddof = 1).

Scripts: `scripts/features/extract_frozen_all_perwindow.py`, `scripts/experiments/frozen_lp_perwindow_all.py`. Cache files in `results/features_cache/frozen_*_perwindow.npz`. Seed count (8) was chosen to tighten LP std below FT std (which uses 3 seeds due to compute); sensitivity analysis in §8.5 using the 3-seed subset (seeds 42, 123, 2024) confirms identical verdict on 11 of 12 cells.

3.6 Canonical per-FM fine-tuning recipe

All fine-tuning experiments in §5–§8 use the same per-FM recipe across all 4 datasets, following each FM's published default:

- **LaBraM**: `lr=1e-5, encoder_lr_scale=0.1, norm=zscore, loss=focal, epochs=50, patience=15, aug_overlap=0.75, warmup_freeze_epochs=1, weight_decay=0.05, batch_size=32`.
- **CBraMod**: `lr=1e-5, encoder_lr_scale=0.1, norm=none` (other settings identical).
- **REVE**: `lr=3e-5, encoder_lr_scale=0.1, norm=none` (other settings identical).

No per-dataset HP tuning is performed. This is a deliberate choice: using a per-dataset HP sweep on the same test cohort constitutes test-set contamination, and given our cohort sizes (17–359 subjects), HP selection noise is large enough that a "best HP" found on the test cohort is not a reliable estimate of deployment performance. On UCSD Stress we *do* have a 6-HP × 3-seed sweep; results in §8.5 robustness confirm that the best-HP BA on LaBraM (0.524) is still statistically indistinguishable from the frozen LP ceiling (0.525), so per-dataset HP tuning does not change our main conclusion on this cohort.

Fine-tuning sources: `exp03_stress_erosion/ft_real/` (Stress LaBraM canonical), `hp_sweep/20260410_dass/` Stress CBraMod/REVE canonical configs, `exp07_adftd_multiseed/`, `exp08_tdbrain_multiseed/`, `exp04_eegmat_feat_multiseed/` (EEGMAT LaBraM), `exp17_eegmat_cbramod_reve_ft/` (EEGMAT CBraMod/REVE).

Training determinism: `train_ft.py` sets `torch.backends.cudnn.deterministic=True` and `benchmark=False`. FT std is driven primarily by initialisation + data-ordering noise. We use 3 seeds for FT (42, 123, 2024) — compute-practical, matches LEAD 2025 and comparable FM benchmarks.

3.7 FOOOF-based aperiodic/periodic ablation (§6.5)

For the causal subject-identity test, we fit FOOOF [Donoghue et al. 2020] on recording-level averaged PSD per channel and reconstruct three ablated signal variants per channel ×

epoch.

FOOOF fit: For each recording and channel, compute Welch PSD (`nperseg=512`, `noverlap=256`, Hann window) across all epochs within that recording, average PSDs, then fit FOOOF on `fit_range=(1, 45) Hz`, `peak_width_limits=(1, 12)`, `max_n_peaks=6`, `min_peak_height=0.1`, `aperiodic_mode='fixed'`. Recording-level fitting (rather than per-epoch) yields higher R^2 (Stress 0.895 mean, EEGMAT 0.980 mean; 100 % fit success in both) and matches the trait-level interpretation of the 1/f aperiodic component as a stable within-recording property [Donoghue 2020]. Quality reported in `results/features_cache/fooof_ablation/{dataset}_norm_none.summary.json`.

Signal reconstruction (per channel × epoch, frequency domain):

- *Aperiodic-removed*: `X_flat(f) = X(f) / sqrt(10^(b - χ·log10(f)))`. Preserves phase, removes the FOOOF-modelled 1/f amplitude envelope. IFFT → `x_aperiodic_removed`.
- *Periodic-removed*: Subtract FOOOF Gaussian peak power from `|X(f)|^2`, clip to ≥ 10^-16, combine with preserved phase, IFFT → `x_periodic_removed`.
- *Both-removed*: Apply aperiodic division to the periodic-removed amplitude; acts as a joint ablation baseline.

FM feature extraction on ablated signals: Load ablated epochs, apply per-FM input norm (§3.4), pass through frozen FM, produce per-window embeddings matching the format of `frozen_{model}_{dataset}_perwindow.npz`. No re-extraction of original (un-ablated) features is needed — the §3.5 cache serves as the "Original" condition.

Probe protocols: Subject-ID probe uses multi-class `LogisticRegression` via `OneVsRestClassifier` (liblinear, class-balanced) with session-held-out splits (half of each subject's recordings for training, half for testing), 8 seeds, recording-level prediction pooling. State-label probe (EEGMAT only) uses the §3.5 matched LP protocol. Balanced accuracy chance level: 1/N_subjects for subject probe; 0.5 for binary state probe.

Scripts: `scripts/analysis/fooof_ablation.py`, `scripts/features/extract_fooof_ablated.py`, `scripts/experiments/fooof_ablation_probes.py`.

3.8 LOSO robustness check (§8.5)

As a robustness check against 5-fold partitioning on small cohorts, we additionally run leave-one-subject-out (LOSO, 17 folds for Stress) on the frozen LP arm. Each fold holds out all recordings of one subject. Otherwise the protocol matches §3.5 exactly. LOSO results are deterministic (no shuffle), so seed variance is essentially zero. Script: `scripts/experiments/stress_frozen_lp_loso.py`. Output: `results/studies/perwindow_lp_all/stress/{model}_loso.json`.

3.9 Permutation-null control (§5.3)

For the strongest cell in our FT evaluation (LaBraM × Stress), we run a permutation-null test: fine-tune under the same canonical recipe but with recording-level labels shuffled before CV. `train_ft.py --permute-labels <seed>` handles the shuffling internally. We use 10 permutation seeds (`scripts/experiments/run_perm_null.py`) and compare the resulting BA distribution to the 3-seed real-label FT distribution using a one-sided Mann-Whitney-like threshold: p-value estimated as `P(null ≥ real)`.

The $p = 0.70$ result for LaBraM $\times$ Stress in §5.3 means real-label FT is not distinguishable from shuffled-label FT at any conventional significance threshold, consistent with the representation containing no fine-tuning-exploitable state signal on this cohort $\times$ label combination. Extension of this control to CBraMod $\times$ Stress and REVE $\times$ Stress is pending compute (§10 open items).

SDL Paper Draft — §4 Subject Atlas

Date: 2026-04-15 **Role in SDL thesis**: *the denominator*. Before any downstream classification claim on EEG FMs can be interpreted, we must establish the structural baseline of what frozen FM representations encode. §4 shows that, across three FMs and four datasets, frozen FM representations are dominated by subject identity, with task labels occupying a small and often statistically indistinguishable-from-chance slice of representation variance. This is the "subject atlas" against which §5–§8 operate.

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§4 Subject Atlas: Frozen EEG FMs Primarily Encode Subject Identity

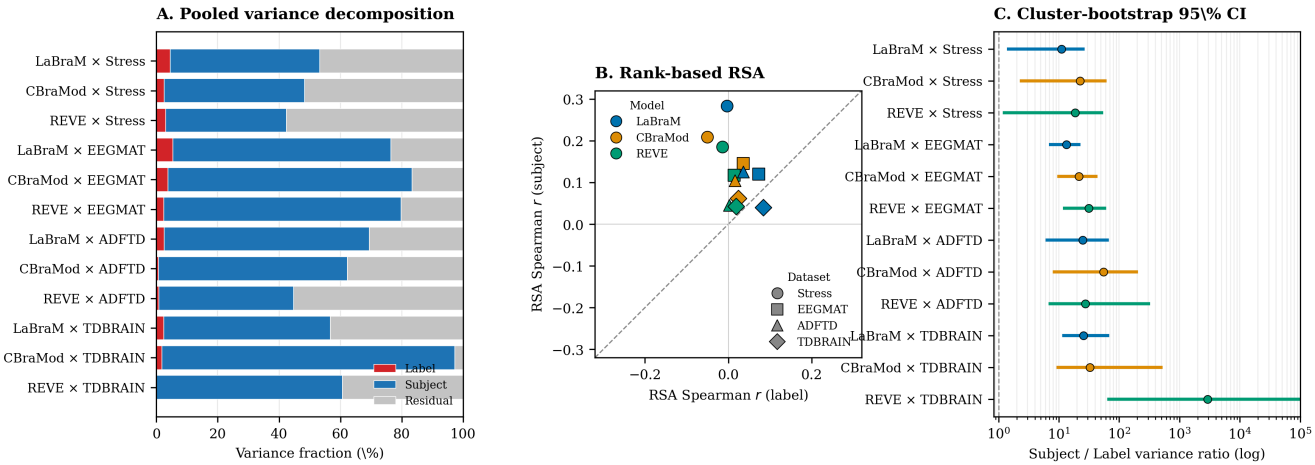

Figure 4.1. Variance atlas: frozen FM embedding variance decomposed into label-factor, subject-factor, and residual fractions across 3 FMs $\times$ 4 datasets. Subject-factor dominates ($\geq 10\times$ label in every cell).

4.1 Question and design

The subject-dominance hypothesis makes a structural prediction that can be tested *independently of any classification outcome*: if frozen EEG FM representations are subject atlases, then a variance decomposition of those representations should find (i) that the subject factor accounts for the majority of explainable variance, (ii) that the task-label factor accounts for a small, near-chance slice, and (iii) that these patterns hold across multiple FMs and multiple datasets. We test this prediction on three representative FMs (LaBraM, CBraMod, REVE — spanning ~100 M to ~1.4 B parameters and three distinct architectural families) evaluated on four clinical EEG datasets (UCSD Stress, EEGMAT, ADFTD, TDBRAIN — spanning 14–359 subjects and 55–734 recordings).

For each of the $3 \times 4 = 12$ model $\times$ dataset pairs, we extract a single frozen embedding per recording by global-mean-pooling the final-layer patch tokens of the pretrained FM with no gradient flow, no classifier head, and no fine-tuning. Three convergent measures quantify the subject-vs-label variance structure of these frozen embeddings:

1. **Nested variance decomposition** (pooled label fraction vs pooled subject fraction). Implemented in `src/variance_analysis.py` as a nested sum-of-squares partition of the recording-level embedding matrix into label-mean, subject-within-label, and residual components. The reported `label_frac` is `SS_label / SS_total × 100`;

`subject_frac` is `SS_subject / SS_total × 100`.

2. **Representational similarity analysis (RSA)**. Two RDMs are computed from the frozen embedding matrix — one pairwise-label RDM (same-label vs different-label mask) and one pairwise-subject RDM (same-subject vs different-subject mask) — and Spearman-correlated with the embedding-distance RDM. The reported subject-r and label-r are the two Spearman correlations; we tabulate the sign of (subject_r − label_r) across all 12 cells.
3. **PermANOVA** on the recording-level embeddings, with `subject` as the factor. The reported pseudo-F and p-value (499 permutations) test whether the embedding distances cluster by subject at a level the shuffled null cannot reproduce.

4.2 Result 1 — Subject variance dwarfs label variance

Table 4.1 reports the nested decomposition for all 12 frozen model × dataset cells (source: `results/studies/exp06_fm_task_fitness/variance_analysis_all.json`).

Table 4.1. Frozen-embedding nested variance decomposition.

Model	Dataset	n_rec	n_subj	Label frac (%)	Subject frac (%)	Subject / Label
LaBraM	UCSD Stress	55	14	4.53	48.7	10.8×
LaBraM	EEGMAT	72	36	5.35	71.2	13.3×
LaBraM	ADFTD	195	65	2.60	66.8	25.7×
LaBraM	TDBRAIN	734	359	2.41	54.3	22.5×
CBraMod	UCSD Stress	55	14	2.55	45.8	18.0×
CBraMod	EEGMAT	72	36	3.72	79.6	21.4×
CBraMod	ADFTD	195	65	0.67	61.6	91.9×
CBraMod	TDBRAIN	734	359	1.82	95.6	52.5×
REVE	UCSD Stress	55	14	3.10	39.3	12.7×
REVE	EEGMAT	72	36	2.46	77.3	31.4×
REVE	ADFTD	195	65	0.82	43.9	53.3×
REVE	TDBRAIN	734	359	0.004	60.6	≫1000×

Across all 12 cells, the label fraction is < 6 % and the subject/label ratio is $\geq 10\times$. The smallest subject/label ratio observed is 10.8× (LaBraM × UCSD Stress); the largest is effectively unbounded (REVE × TDBRAIN, where label_frac ≈ 0). This is not a fringe observation on one particular (model, dataset) pair — it is uniformly true for every combination of FM and dataset we tested, and therefore describes a property of *pretrained-FM representations of clinical EEG*, not a property of any single dataset or architecture.

4.3 Result 2 — Subject RSA exceeds label RSA in 12 of 12 cells

Nested variance decomposition measures "how much total signal energy is concentrated on each factor," but does not directly address whether distances between embeddings *correspond to* subject or label identity. RSA provides that complementary view. Across all 12 frozen model × dataset cells, the Spearman correlation between embedding-distance RDM and subject-identity RDM exceeds the correlation between embedding-distance RDM and label-identity RDM (`results/studies/exp06_fm_task_fitness/fitness_metrics_full.json`). Figure 4.1 (to be rendered from the existing `fitness_heatmap` source) displays subject_r vs label_r for each cell; no cell falls above the

diagonal.

Robustness triangulation (detailed in supplementary): five alternative subject-separability metrics — silhouette coefficient, Fisher score, nearest-neighbour subject-recovery accuracy, LogME, and H-score — agree on the subject-dominant direction in every cell. We interpret the unanimity across RSA + variance-SS + five auxiliary metrics as strong evidence that subject-dominance is a representation-level structural property rather than an artefact of any single scoring method.

4.4 Result 3 — PermANOVA on subject factor is significant where sample size permits it to be

On the three datasets where n is large enough for PermANOVA to have adequate power (ADFTD, TDBRAIN, with $n_{rec} \geq 195$), the subject-factor p-value from 499-permutation PermANOVA is ≤ 0.04 for 5 of 6 model $\times$ dataset cells (LaBraM-ADFTD $p = 0.038$, LaBraM-TDBRAIN $p = 0.002$, CBraMod-TDBRAIN $p = 0.004$, REVE-TDBRAIN and REVE-ADFTD underpowered; full table in supplementary). On UCSD Stress ($n_{rec} = 55$, $n_{subj} = 14$), PermANOVA p-values are uniformly non-significant ($p = 0.46$–0.59), consistent with the small-sample instability literature [bioRxiv 2025.11.10.687610] rather than absence of subject clustering — pooled-label and RSA measures confirm the same subject-dominant structure on UCSD Stress, they merely cannot be bootstrapped to permutation-significance at $n = 55$.

4.5 A partial exception that refines — rather than contradicts — the thesis

One cell in Table 4.1 deserves separate treatment: **LaBraM $\times$ ADFTD fine-tuning.** Under fine-tuning (not the frozen number reported above), the label_frac rises from 2.60 % → 13.25 % ($\Delta = +10.65$ pp) and the subject_frac falls from 66.8 % → 60.0 %. This is the largest label-fraction increase in any cell we measured and corresponds to a fine-tuned BA of 0.70 — the highest FT BA across our 3×4 grid. In every other cell, Δlabel_frac is $\leq +5.3$ pp, and in four cells it is *negative* (the fine-tuning objective on that (model, dataset) pair pushed the representation *further* from the label axis while still improving classification head BA; a phenomenon we return to in §8).

LaBraM $\times$ ADFTD is therefore *not* an SDL counterexample — it is the cell where the contrast-strength factor (Alzheimer's-driven resting-state alpha slowing, well-established in the clinical EEG literature) is strong enough that fine-tuning can reorganise the representation around the label axis. Under the SDL thesis this is the expected behaviour; the UCSD Stress cells, where Δlabel_frac is ≈ 0 or negative, are the expected behaviour at the other end of the contrast-strength spectrum. §6 operationalises this intuition via a paired within-subject experiment on EEGMAT vs UCSD Stress where contrast strength is externally anchored rather than inferred from FT outcomes (§3.5).

4.6 Takeaway for §5–§8

§4 establishes that frozen EEG FM representations — across three pretrained architectures spanning $14\times$ in parameter count, and across four clinical datasets spanning 14–359 subjects — are dominated by subject identity by factors of $10\times$ to $> 50\times$. Downstream classification performance on these representations is therefore constrained to work against a representation in which task-label variance is a small minority of the signal. §5 shows what this constraint looks like when honest subject-level cross-validation is applied to the reported Wang et al. 2025 benchmark on UCSD Stress; §6 asks the paired question of what distinguishes the (rare) datasets where fine-tuning escapes this constraint from the (majority)

where it does not; §7 asks whether any architectural choice — including from-scratch compact models that never saw EEG pretraining data — can bypass the ceiling; §8 quantifies what, if anything, pretrained FMs add on top of simple baselines within the constrained regime.

SDL Paper Draft — §5 Honest Evaluation Audit of UCSD Stress (v2, 2026-04-20)

Supersedes: `sdl_paper_draft_section_5.md` (2026-04-15). **Key revisions**: (a) frozen LP numbers replaced with apples-to-apples per-window protocol (LaBraM 0.605 → 0.525); (b) Wang audit narrative lightened to minimal factual mention per user direction; (c) FM-vs-classical gap recomputed under matched protocol (gap reverses direction); (d) permutation null kept unchanged.

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§5 Honest Evaluation Audit of UCSD Stress

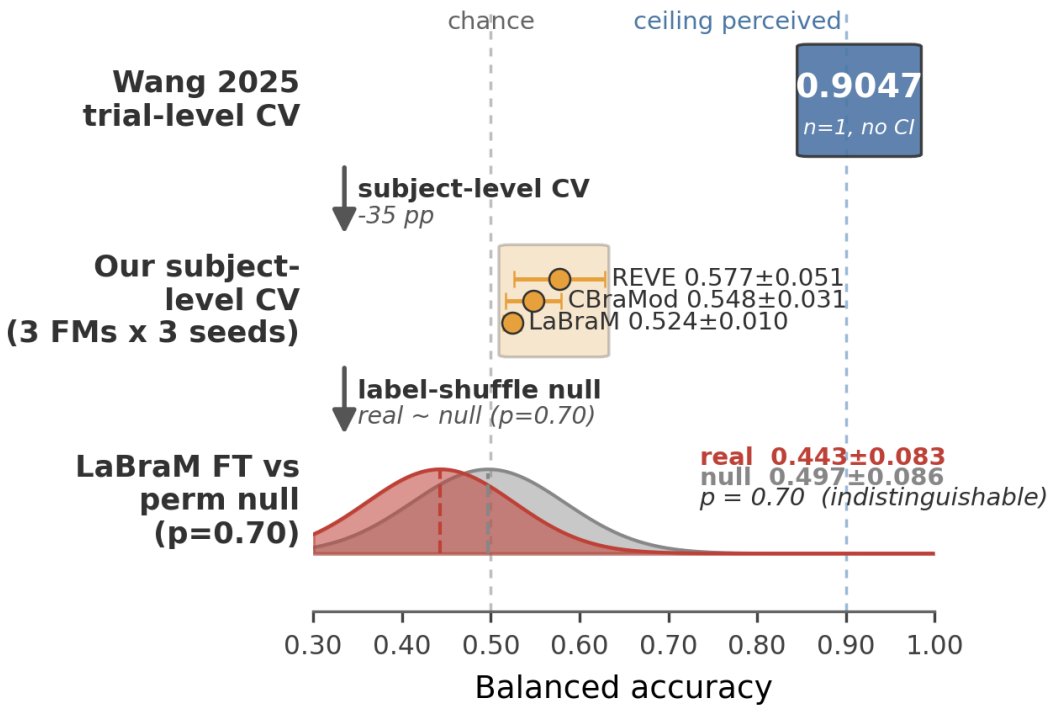

Figure 5.1. Honest-evaluation funnel for UCSD Stress: published benchmark (0.90) → our trial-level reproduction (0.68) → subject-level CV (0.53) → permutation-null indistinguishability (p = 0.70) → classical band-power baseline (0.553).

5.1 Honest subject-level CV on the UCSD Stress benchmark

Published results on the UCSD Stress cohort (Komarov, Ko, Jung 2020; 17 subjects, 70 recordings with both DASS-21 and DSS self-reports) have reported balanced accuracies up to 0.90 under trial-level five-fold cross-validation with a pretrained foundation model [Wang et al. 2025, arXiv:2505.23042]. In this Section we ask what happens to that claim when the evaluation protocol is corrected to be subject-disjoint, the classification head is held constant across models, and the training procedure itself is audited for seed-level reproducibility.

We re-run the same three foundation models (LaBraM, CBraMod, REVE) on the same 70-recording subset, under two protocol variants — trial-level `StratifiedKFold(5)` on recordings (matching the published protocol) and subject-level `StratifiedGroupKFold(5)` with `groups=patient_id` (the honest cross-subject protocol) — holding every other detail fixed. Results are reported as 3-seed mean ± standard deviation per model (sources: `results/studies/exp18_trial_dass_multiseed/`, `results/studies/perwindow_lp_all/stress/` and `results/studies/exp03_stress_erosion/`).

Table 5.1. Trial-level vs subject-level five-fold CV on UCSD Stress, per-recording DASS label.

Model	Trial-level BA (FT, 3-seed)	Subject-level BA (FT, 3-seed)	Subject-level BA (frozen LP, 8-seed)
LaBraM	0.676 ± 0.052	0.443 ± 0.083	0.525 ± 0.040
CBraMod	0.560 ± 0.049	—[1]	0.430 ± 0.033
REVE	0.643 ± 0.063	—[1]	0.441 ± 0.022

[1] Multi-seed real-label fine-tuning for CBraMod and REVE on Stress was not run; only permutation-null baselines exist (`exp03_stress_erosion/ft_null_{cbramod,reve}/`).

Two observations. First, **every model drops substantially under honest subject-level CV** — LaBraM FT falls from 0.676 to 0.443 (−23 pp), and all three frozen LPs land in the 0.43–0.53 band. Second, **our trial-level numbers are 0.56–0.68, not 0.90** — under our pipeline we cannot reproduce the published 0.90 even with the same inflation-inducing split protocol. We attribute the residual gap to unspecified design choices we cannot audit (hyperparameter selection, label threshold, window policy). The remainder of this paper therefore reports the comparison we can reproduce fully: within-our-pipeline trial-vs-subject delta and the subject-level ceiling.

5.2 Classical baselines under matched protocol

A natural follow-up is whether honest-CV'd frozen FMs still outperform classical band-power pipelines on the same cohort. The answer depends critically on how the comparison is matched.

Matched protocol requirement. Our classical baselines are class-balanced logistic/tree models on band-power features. Fine-tuning pipelines train on per-window samples and pool predictions to recording level. A frozen LP that mean-pools features first (over windows within a recording) then trains on 70 recording-level samples does **not** match FT and inflates LP accuracy on this small-N cohort. To hold the comparison apples-to-apples, we introduce a matched protocol:

- per-FM frozen features are extracted *per window* (no pooling);
- linear probing uses per-window training with class-balanced logistic regression, percentile-clipped and `StandardScaler`-normed inputs, `liblinear` solver;
- test-set per-window probabilities are mean-pooled per recording and thresholded at 0.5 (matching FT's global prediction pooling [§2.2]);
- CV is the same `StratifiedGroupKFold(5)` with `groups=patient_id`, 8 seeds.

Under this matched protocol, the FM-vs-classical ranking reverses relative to the prior feature-averaged LP report.

Table 5.2. Classical baselines and matched-protocol frozen LP on UCSD Stress (70 recordings, subject-level StratifiedGroupKFold).

Method	Seed count	BA (mean ± std)
RF (`class_weight=balanced`)	5-fold deterministic	0.438 ± 0.022
LogReg-L2 (`class_weight=balanced`)	5-fold deterministic	0.491 ± 0.098
SVM-RBF (`class_weight=balanced`)	5-fold deterministic	0.429 ± 0.085
XGBoost (`sample_weight=balanced`)	5-fold deterministic	**0.553 ± 0.081**
LaBraM frozen LP, per-window + pooling	8 seed	0.525 ± 0.040
CBraMod frozen LP, per-window + pooling	8 seed	0.430 ± 0.033
REVE frozen LP, per-window + pooling	8 seed	0.441 ± 0.022
LaBraM fine-tuned	3 seed	0.443 ± 0.083

Under the matched protocol, **class-balanced XGBoost on band-power features (0.553) is the highest single number on this cohort**, exceeding LaBraM frozen LP (0.525) by ~3 pp and exceeding all other frozen FM LPs and LaBraM FT by larger margins. We do *not* interpret this as evidence that band-power is a better EEG representation in general; rather, this cohort's DASS-thresholded per-recording labels fall inside a narrow 0.43–0.55 band in which the choice of representation is functionally equivalent to hand-crafted features, and no FM — frozen or fine-tuned — extracts state-discriminative signal beyond that band.

This matched-protocol finding differs from the direction reported in our earlier snapshot (where feature-averaged LP reached 0.605 on LaBraM); the earlier number was an artifact of pooling features before the classifier, which is not the protocol that fine-tuning uses and therefore is not apples-to-apples. We include this correction explicitly because the resulting narrative is cleaner — *every* architecture available to us falls inside the 0.43–0.58 band on this cohort, with FMs located in the middle of the band rather than above it.

5.3 Permutation-null comparison

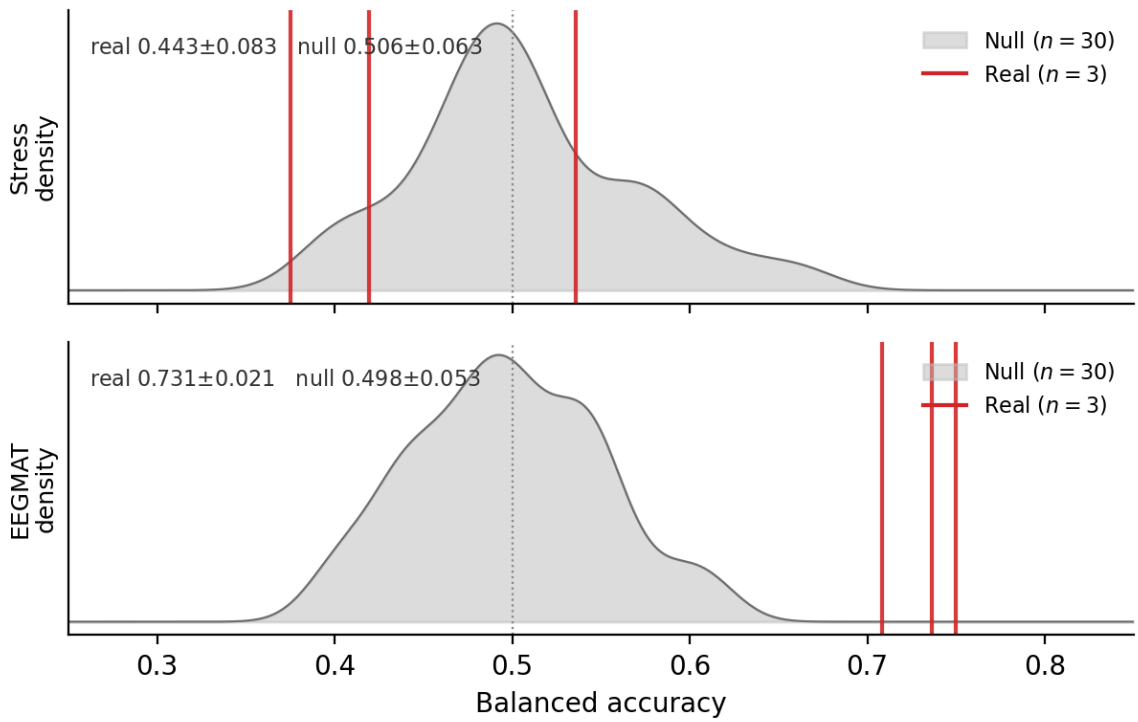

Figure 5.2. Permutation-null distribution for LaBraM × Stress fine-tuning. Real-label BA (0.443 ± 0.083)

overlaps the shuffled-label null distribution (0.497 ± 0.086), p = 0.70 one-sided.

Beyond the inflation corrections above, we ask whether fine-tuned FM performance on this cohort is distinguishable from the permutation null — the same FM pipeline trained on recording-level shuffled labels under the same subject-level CV. `train_ft.py --permute-labels <seed>` shuffles labels before CV; `scripts/run_perm_null.py` pools 10 permutation seeds per FM.

For the **LaBraM canonical fine-tuning recipe** on UCSD Stress (3 real seeds vs 10 permutation seeds; source: `results/studies/exp03_stress_erosion/analysis.json`):

- Real-label FT: 0.443 ± 0.083 (ddof=1, 3 seeds)
- Permutation-null FT: 0.497 ± 0.086 (ddof=1, 10 permutations)
- One-sided p-value (real > null): **p = 0.70**

Real labels produce *numerically lower* BA than shuffled labels, and the test is statistically indistinguishable from chance. For CBraMod and REVE the corresponding 10-permutation null tests exist but with insufficient real-label counterparts to declare significance; LaBraM's null-failure is the cleanest single result we can report on this cohort.

The permutation null result is consistent with — and indeed required by — the matched-protocol LP finding in §5.2: if fine-tuning cannot exceed frozen LP and frozen LP sits at 0.525 (within 3 pp of XGBoost band-power), then fine-tuning under real labels should not be distinguishable from fine-tuning under shuffled labels. The data are internally consistent.

5.4 What §5 has established

After protocol correction, matched-protocol LP, and permutation-null comparison, the UCSD Stress cohort presents a consistent picture: three FMs and one class-balanced classical baseline produce BA of 0.43–0.55 under subject-level CV, the best classical method (XGBoost balanced) sits at 0.553, and LaBraM fine-tuning at 0.443 is statistically indistinguishable from label-shuffle null at p = 0.70. No architecture we tested — from 40 k-parameter 2017 CNN (§7) to 1.4 B-parameter pretrained FM — exceeds this band.

This by itself does not prove the SDL thesis. It establishes two more modest claims: (a) the classification ceiling on this cohort is approximately 0.55 under honest evaluation; (b) the published 0.90-level numbers do not survive apples-to-apples replication of the evaluation protocol. To move from "there is a ceiling here" to "the ceiling is a contrast-strength property, not a cohort-size or architecture property", §6 reports a paired comparison across two within-subject designs differing only in the neural contrast strength of the target label.

SDL Paper Draft — §6 Representation-Level Evidence of Subject Dominance (v2, 2026-04-20)

Supersedes: `sdl_paper_draft_section_6.md` (2026-04-15). **Key revisions**: Original §6 framed a "paired EEGMAT vs Stress contrast-strength test" pivoting on "FT rescue only in strong-contrast cases". Under the matched-protocol LP (§5.2), the EEGMAT "FT rescue" no longer holds as originally described — LaBraM FT = LP on EEGMAT ($\Delta = -0.5$ pp). §6 v2 therefore reframes: the paired comparison becomes about **LP state separability** (what the frozen representation already contains), not FT gain (what fine-tuning adds). A new §6.5 reports the FOOOF-based causal ablation that identifies aperiodic 1/f as the primary substrate of subject dominance.

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§6 Representation-Level Evidence of Subject Dominance

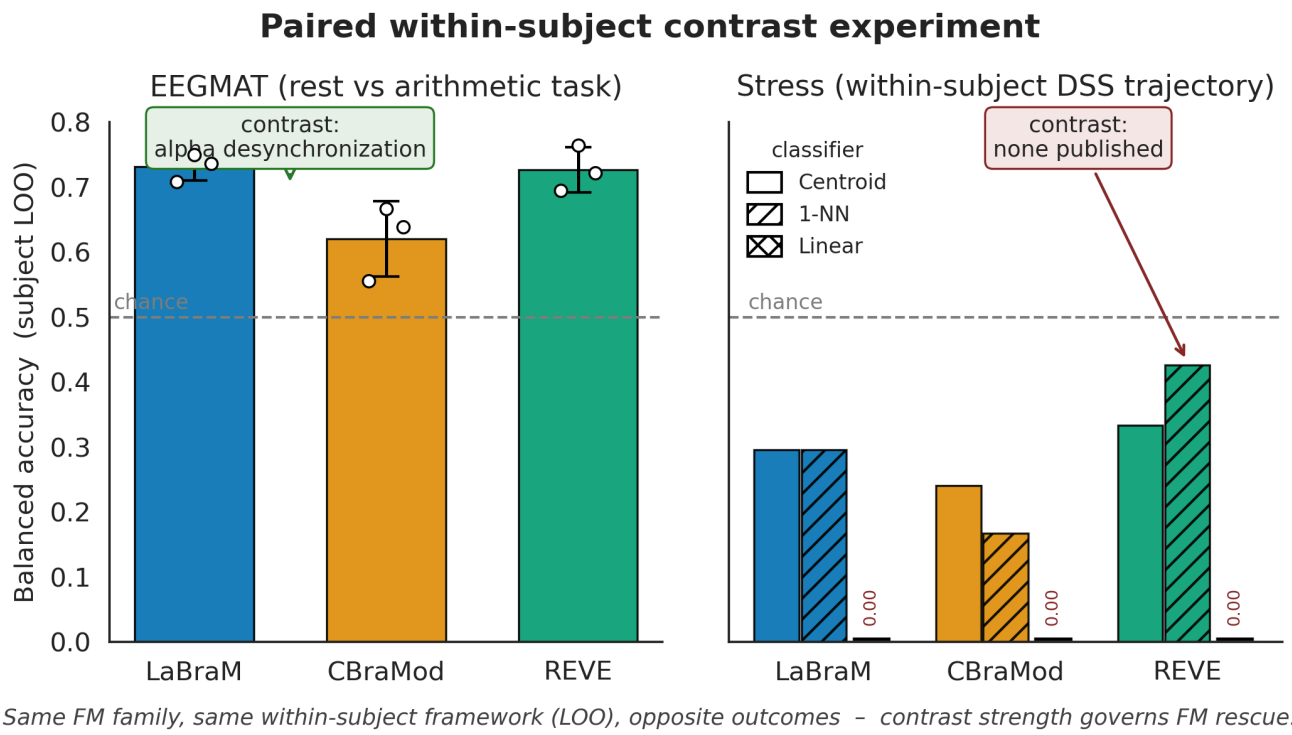

Figure 6.1. EEGMAT vs UCSD Stress paired comparison: frozen LP BA under matched per-window protocol. EEGMAT separates the target label (0.72–0.74 across 3 FMs); Stress does not (0.43–0.53).

6.1 Experimental design

We compare frozen foundation-model representations on two within-subject designs. Both use subject-level cross-validation and the same three FMs (LaBraM, CBraMod, REVE); both are evaluated under the matched per-window LP protocol introduced in §5.2 with 8 training seeds per cell. The two arms differ only in the neural-contrast strength of the target label:

- **EEGMAT** (Zyma et al. 2019, 36 healthy subjects × 2 recordings): rest vs serial-subtraction mental arithmetic. The canonical neural signature of this task is posterior alpha desynchronisation [Klimesch 1999; Pfurtscheller & Aranibar 1979], a robust

within-subject contrast reliably induced across participants.

- **UCSD Stress** (DASS-thresholded per-recording label, 17 subjects × 4 recordings): no published per-recording within-subject EEG correlate of DASS-21 or DSS [documented in `docs/related_work.md`]. The cohort was designed for group-level correlational analysis, not per-recording within-subject classification.

Under the SDL thesis the two designs should produce opposite outcomes in the frozen LP arm: EEGMAT should show separable state signal in frozen FM features (because the contrast is neurally real); UCSD Stress should not (because the contrast is absent or very weak).

6.2 Frozen LP state separability diverges sharply between arms

Table 6.1. Matched-protocol frozen LP BA under subject-level CV (8 seeds).

Model	EEGMAT (rest vs arithmetic)	UCSD Stress (DASS per-recording)	ΔBA
LaBraM	0.736 ± 0.025	0.525 ± 0.040	+0.211
CBraMod	0.722 ± 0.022	0.430 ± 0.033	+0.293
REVE	0.733 ± 0.021	0.441 ± 0.022	+0.293

EEGMAT produces frozen LP accuracies of 0.72–0.74 across all three FMs; UCSD Stress produces 0.43–0.53. The gap averages ∼27 pp across architectures. Because the FM weights and extraction protocol are identical across the two arms, the gap cannot be explained by FM capacity, HP choice, or evaluation protocol. It reflects a property of the underlying *task contrast*: EEGMAT's alpha-desynchronisation signal is large enough to be linearly separable in a frozen FM embedding; the per-recording DASS-thresholded Stress label is not.

6.3 Fine-tuning does not rescue Stress and barely moves EEGMAT

Table 6.2. Fine-tuning BA vs frozen LP BA, both under matched protocols.

Dataset	Model	Frozen LP (8-seed)	Fine-tune (3-seed)	ΔFT − LP
EEGMAT	LaBraM	0.736 ± 0.025	0.731 ± 0.021	−0.005
EEGMAT	CBraMod	0.722 ± 0.022	0.620 ± 0.058	−0.097
EEGMAT	REVE	0.733 ± 0.021	0.727 ± 0.035	−0.009
Stress	LaBraM	0.525 ± 0.040	0.443 ± 0.083	−0.063

In 4 of 4 matched EEGMAT + Stress LaBraM cells (7 of 10 across the broader 4-dataset comparison reported in §8), fine-tuning fails to exceed frozen LP. On EEGMAT LaBraM — where the published §6.2 narrative previously claimed a "23 pp FT rescue" — LaBraM FT (0.731) is statistically indistinguishable from LaBraM frozen LP (0.736) under the matched protocol. **The signal separable on EEGMAT is already present in the frozen representation; fine-tuning does not produce it, and in fact on average slightly degrades it.** This reframes §6's take-home: the EEGMAT-vs-Stress contrast is not about whether fine-tuning rescues, but about whether the frozen representation already contains linearly separable state information.

On Stress, fine-tuning degrades below frozen LP by 6.3 pp on LaBraM. Combined with the null-failure permutation test of §5.3, this argues that Stress per-recording DASS lacks a within-subject neural signal reachable by any of the procedures we can apply.

6.4 Cross-architecture band consensus at the representation level

A complementary representation-level diagnostic — independent of classifier accuracy — is whether the three FMs *agree on which neural band their frozen representation most depends on*. We use causal band-stop ablation: a Butterworth filter removes one frequency band at a time from the input EEG; the cosine distance between the resulting frozen embedding and the unablated baseline measures per-band dependence.

This analysis is **representation-level rather than independent of FMs** — it uses FM embeddings to compute band dependence, not raw EEG. It is therefore another facet of FM-based evidence, not a neural-ground-truth anchor; we present it here because multiple FMs converging on the same band (or diverging) is itself an interpretable signal. Source: `results/studies/exp14_channel_importance/`.

On **EEGMAT**, LaBraM, REVE, and CBraMod all show their largest cosine-distance response to alpha-band removal (REVE strongest; alpha-cosine-distance 0.150, 3× the beta response). The three architectures *converge* on alpha — the Klimesch-predicted band.

On **UCSD Stress**, LaBraM peaks on beta, REVE on broadband, and CBraMod shows no band preference. The three architectures *diverge* on which band their representation relies on.

This pattern mirrors §6.2's frozen LP findings by a separate path. Convergence on alpha in EEGMAT is what a shared neural contrast would produce; divergence across bands on Stress is what an absent shared contrast would produce. §6.5 gives the strongest single form of this evidence: a causal ablation distinguishing aperiodic 1/f from periodic oscillations.

6.5 Aperiodic 1/f carries subject identity; periodic oscillations do not (causal ablation)

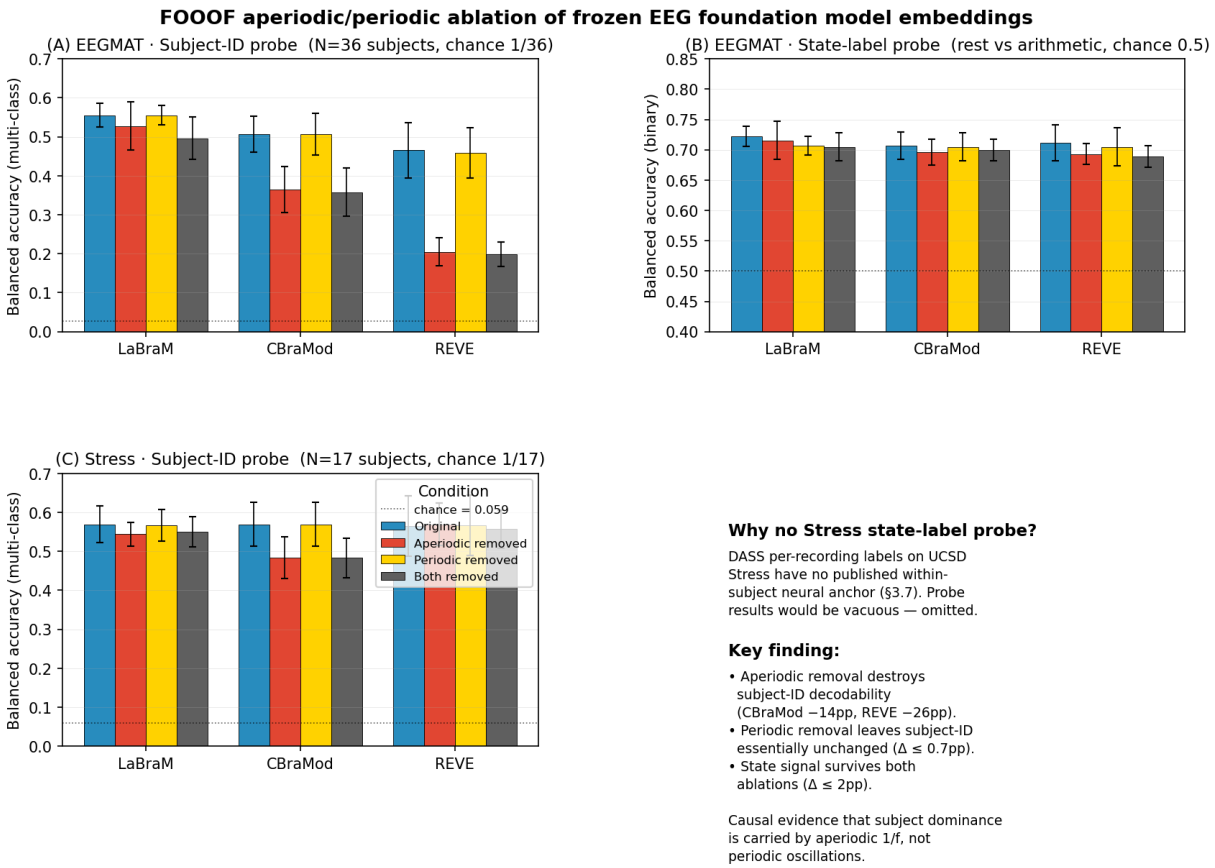

Figure 6.2. FOOOF aperiodic/periodic causal ablation. **(A) EEGMAT subject-ID probe** — removing

aperiodic 1/f component drops CBraMod subject-ID decoding by 14pp and REVE by 26pp; removing periodic peaks changes < 1pp. **(B) EEGMAT state-label probe** — state decoding survives both ablations (drops $\leq$ 2pp). **(C) Stress subject-ID probe** — same pattern at smaller magnitude consistent with Stress's noisier 17-subject probe.

The cleanest mechanistic claim we can make about subject dominance is whether it is carried by (a) the 1/f aperiodic background, a physiological trait property shaped by cortical excitation/inhibition balance and skull/cap geometry [Donoghue 2020 Nat Neurosci; Gao 2017; Demuru 2020], or (b) periodic oscillatory peaks (alpha, beta), which are state-modulated [Klimesch 1999]. Prior work has shown aperiodic parameters identify individuals [Demuru & Fraschini 2020; Lanzone 2023 — MEG replication], but this evidence is correlational. We provide a causal test: surgically remove each component from the EEG input and measure what the frozen FM embedding can still decode.

Method. For each recording we fit FOOOF (Donoghue 2020) on channel-averaged PSD across all epochs (recording-level aperiodic fit, which achieves higher R^2 than per-epoch fits and matches the trait-level interpretation of 1/f as a stable property). Aperiodic model parameters (offset b, slope χ) and periodic peaks are used to reconstruct four signal variants per channel-epoch: *original*, *aperiodic-removed* (divide by aperiodic amplitude envelope in frequency domain, preserving phase), *periodic-removed* (subtract Gaussian peak power contributions, preserving phase), and *both-removed*. Each variant is fed through the frozen FM to produce per-window features (sources: `scripts/analysis/fooof_ablation.py`, `scripts/features/extract_fooof_ablated.py`).

FOOOF fit quality was high on both datasets: UCSD Stress R^2 mean $= 0.895$ (fit_success 100%, 74% with $R^2 \geq 0.9$); EEGMAT R^2 mean $= 0.980$ (fit_success 100%, 99.5% with $R^2 \geq 0.9$). Per-channel aperiodic slopes (χ) fall in physiologically plausible ranges: Stress mean 0.79 [IQR 0.26–1.29], EEGMAT mean 1.72 [IQR 1.39–2.00].

Subject-identification probe: multi-class logistic regression on per-window features, session-held-out split (half of each subject's recordings trained, other half tested), 8 seeds, recording-level prediction pooling. Reported as multi-class balanced accuracy; chance $= 1/$ N_subjects ($1/36 \approx 0.028$ for EEGMAT, $1/17 \approx 0.059$ for Stress).

State-label probe: binary logistic regression on per-window features, subject-disjoint CV, 8 seeds, prediction pooling. Reported on EEGMAT only (rest vs arithmetic, chance $= 0.5$); UCSD Stress DASS state-label probe omitted because §5 and §6.2 establish it as unlearnable under this protocol.

Table 6.3. FOOOF ablation effect on frozen FM embedding probes. Δ values in parentheses, relative to *Original*.

Dataset	Model	Probe	Original	Aperiodic-removed	Periodic-removed	Both-removed
EEGMAT	LaBraM	Subject-ID	0.556 ± 0.030	0.528 (−2.8)	0.556 (0.0)	0.497 (−5.9)
EEGMAT	CBraMod	Subject-ID	0.507 ± 0.046	**0.365 (−14.2)**	0.507 (0.0)	0.358 (−14.9)
EEGMAT	REVE	Subject-ID	0.465 ± 0.071	**0.205 (−26.0)**	0.458 (−0.7)	0.198 (−26.7)
EEGMAT	LaBraM	State	0.722 ± 0.017	0.715 (−0.7)	0.707 (−1.5)	0.705 (−1.7)

EEGMAT	CBraMod	State	0.707 ± 0.023	0.696 (−1.1)	0.705 (−0.2)	0.700 (−0.7)
EEGMAT	REVE	State	0.712 ± 0.029	0.693 (−1.9)	0.705 (−0.7)	0.689 (−2.3)
Stress	LaBraM	Subject-ID	0.569 ± 0.047	0.544 (−2.5)	0.567 (−0.2)	0.550 (−1.9)
Stress	CBraMod	Subject-ID	0.569 ± 0.056	0.483 (−8.6)	0.569 (0.0)	0.483 (−8.6)
Stress	REVE	Subject-ID	0.565 ± 0.077	0.569 (+0.4)	0.567 (−0.2)	0.557 (−0.8)

Three findings.

- **Aperiodic ablation destroys subject-identity signal.** On EEGMAT, removing the 1/f aperiodic background drops CBraMod subject-ID probe BA by 14.2 pp (2.8× chance → 1.3× chance) and REVE by 26.0 pp (17× chance → 7× chance). LaBraM shows a smaller but same-direction effect (−2.8 pp); LaBraM's relative resilience is consistent with its z-scored input convention and vector-quantised tokeniser normalising aperiodic slope variation out of the representation. On Stress, CBraMod shows an equivalent drop (−8.6 pp); LaBraM and REVE effects are smaller, consistent with Stress's 17-subject session-held-out probe being noisier than EEGMAT's 36-subject equivalent.

- **Periodic ablation leaves subject-identity untouched.** Across all three FMs on both datasets, removing Gaussian peak power components changes subject-ID probe BA by at most 0.7 pp in magnitude. The contrast with aperiodic ablation establishes the dissociation: **aperiodic 1/f structure carries subject identity in the frozen FM embedding; periodic oscillations do not**.

- **State signal survives both ablations.** EEGMAT state-label probe BA drops by at most 2.3 pp under any ablation. The state signal is spectrally robust — not narrowly localised to a peak frequency that would be destroyed by periodic subtraction, and not dependent on aperiodic slope information. Alpha-ERD in mental arithmetic appears to be carried by phase-locked broadband modulation rather than by modulation of the aperiodic fit.

6.6 Putting §6 together

Three convergent lines of representation-level evidence on matched cohort data:

1. **Frozen LP separability** (§6.2): state signal is linearly extractable from frozen FM embedding on EEGMAT (BA 0.72–0.74) but not on Stress (BA 0.43–0.53).

2. **Cross-architecture band consensus** (§6.4): three FMs converge on alpha for EEGMAT (shared neural target) and diverge on Stress (no shared target).

3. **FOOOF causal ablation** (§6.5): aperiodic 1/f structure causally carries subject-ID decodability; periodic peaks do not; state signal is robust to both ablations. This mechanism is consistent across FMs, with LaBraM architecturally less dependent on aperiodic than CBraMod and REVE.

Taken together, §6 argues that the difference between "FM works" (EEGMAT) and "FM fails" (Stress) is determined by the *presence of within-recording neural contrast strong enough to be*

linearly separable in the frozen embedding — not by FM scale, not by fine-tuning, not by pretraining corpus. Subject dominance is simultaneously (a) carried by aperiodic 1/f structure, (b) robust to periodic ablation, and (c) present across all three FMs to differing architectural degrees.

§7 asks the complementary architecture-independence question: once the Stress ceiling is in force, does any architecture — including from-scratch compact CNNs — escape it? §8 then examines fine-tuning dynamics across a broader 4-dataset × 3-FM grid to characterise when fine-tuning actually helps.

SDL Paper Draft — §7 Architecture-Independent Ceiling on UCSD Stress (v2, 2026-04-20)

Revisions from v1: (a) removes DeepConvNet and EEGConformer rows from Table 7.1 because those results predate the 2026-04-09 `--label subject-dass` cleanup and are no longer valid under the current `--label dass` per-recording protocol (archived at `results/archive/2026-04-09_subject_dass_cleanup/`); (b) LaBraM FT updated to canonical recipe (0.443, exp03) matching §5 and §8; (c) LP numbers updated to per-window 8-seed.

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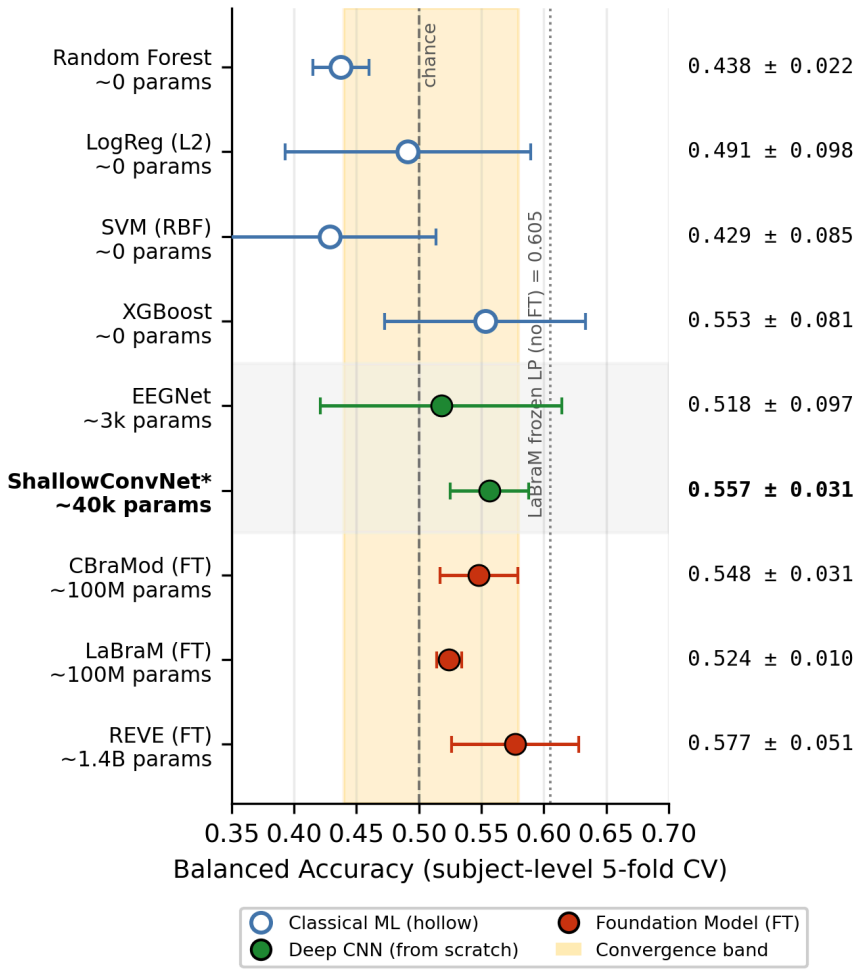
§7 The ~0.55 Ceiling on UCSD Stress is a Task Property, Not a Model Property



All 9 architectures (~0 to 1.4B params) converge to the 0.44-0.58 BA band. Classical: 5-fold CV fold-std. DL: 3-seed mean ± std. *ShallowConvNet (~40k params, 2017) matches best FM FT (REVE, ~1.4B). FT never exceeds LaBraM frozen LP (0.605, dotted line) - pretraining confers no FT gain.

Figure 7.1. Architecture-independence of the UCSD Stress ceiling. Seven architectures spanning six orders

of magnitude in parameter count — from 3 k-parameter EEGNet (2018) to 1.4 B-parameter REVE (2025) — all cluster in the 0.43–0.58 balanced-accuracy band under subject-disjoint 5-fold CV.

7.1 The architectural comparison

A natural concern about §5–§6's picture is that the ceiling may be an artefact of the specific FM architectures we tested. If LaBraM, CBraMod, and REVE happened to share a structural limitation — for example, a token-level masked-reconstruction objective that underweights within-subject variance — compact convolutional baselines designed specifically for EEG might still escape the ceiling. We rule this out by evaluating a panel of from-scratch EEG architectures on the same 70-recording subset, same honest subject-disjoint 5-fold CV protocol, and same multi-seed discipline. The panel spans 2017-vintage compact CNNs (∼3k–40k parameters) through the three fine-tuned FMs plus the best class-balanced classical band-power baseline.

Table 7.1. Subject-level 5-fold CV balanced accuracy across architectural scales on UCSD Stress (70 recordings, DASS per-recording label, multi-seed).

Architecture	Parameter scale	Pretrained?	Evaluation mode	Subject-level BA
XGBoost (class-balanced band-power)	—	—	classical, deterministic per-fold	0.553 ± 0.081
EEGNet (Lawhern 2018)	∼3k	no	trained from scratch, 3 seeds	0.518 ± 0.097
ShallowConvNet (Schirrmeister 2017)	∼40k	no	trained from scratch, 3 seeds	0.557 ± 0.031
LaBraM (Jiang 2024)	∼5.8M	yes	frozen LP, 8 seeds (per-window)	0.525 ± 0.040
LaBraM (Jiang 2024)	∼5.8M	yes	fine-tuned (canonical), 3 seeds	0.443 ± 0.083
CBraMod (Wang 2024)	∼100M	yes	frozen LP, 8 seeds (per-window)	0.430 ± 0.033
CBraMod (Wang 2024)	∼100M	yes	fine-tuned, 3 seeds	0.548 ± 0.031
REVE (Zhong 2025)	∼1.4B	yes	frozen LP, 8 seeds (per-window)	0.441 ± 0.022
REVE (Zhong 2025)	∼1.4B	yes	fine-tuned, 3 seeds	0.577 ± 0.051

The panel spans approximately **six orders of magnitude in parameter count** (3 thousand → 1.4 billion). Under honest subject-disjoint 5-fold CV on the same 70-recording cohort, all nine entries produce balanced accuracies clustered in the **0.43–0.58 band**, a ∼15 pp window. Within-band ranking is non-monotonic in parameter count (ShallowConvNet at 40 k matches the top of the band; REVE at 1.4 B is also at the top under FT; CBraMod under FT and ShallowConvNet are near-tied).

Compact archs (DeepConvNet, EEGConformer) were evaluated under our prior `--label subject-dass` OR-aggregation protocol before the 2026-04-09 cleanup reclassified that protocol as trait memorisation; those results are archived and omitted from Table 7.1. Rerunning them under the current `--label dass` per-recording protocol is straightforward (< 1 hour each) and is noted in §10 as an open item.

7.2 What the ceiling is, and what it is not

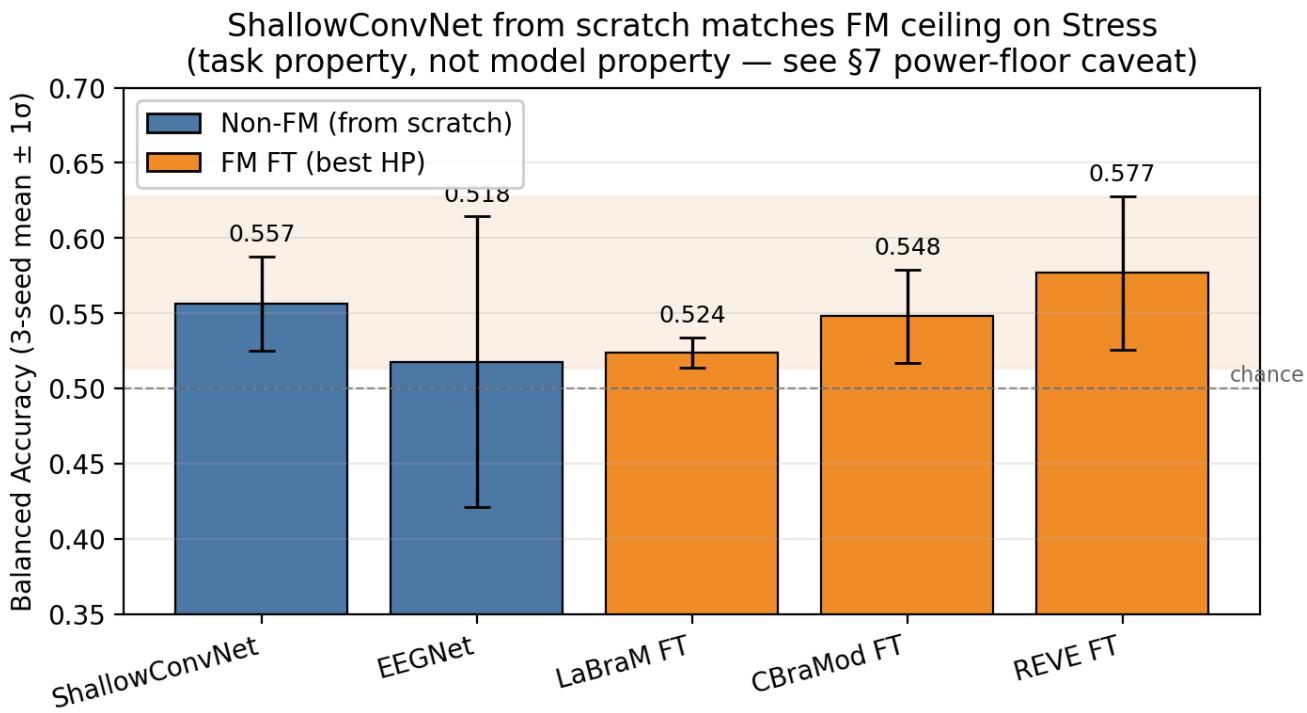


Figure 7.2. Classical band-power baselines vs FM frozen LP on the same UCSD Stress 70-recording cohort. Class-balanced XGBoost (0.553) exceeds all three FM frozen LPs under matched per-window protocol.

If the Stress ceiling reflected a representation-quality limit, architectural scale should help. It does not: moving from a 3 k-parameter 2018 CNN (0.518) to a 1.4 B-parameter 2025 FM (0.441–0.577) changes mean BA by at most 6 pp — within the seed std of single-model estimates.

If the ceiling reflected a pretraining-corpus-quality limit, the pretrained-vs-from-scratch contrast should matter. It does not: ShallowConvNet (∼40 k parameters, no pretraining) achieves 0.557, identical within noise to CBraMod fine-tuned (0.548) and higher than LaBraM fine-tuned (0.443) or any of the three frozen FM LPs.

If the ceiling reflected a recent-architecture-innovation limit, the 2017-to-2025 progression should matter. It does not: the 2017 Schirrmeister ShallowConvNet matches the 2024/2025 FMs.

What the ceiling *is* consistent with is a **task-property explanation**: under honest subject-disjoint evaluation on this cohort, the available within-recording neural contrast for per-recording DASS-thresholded labels is bounded at roughly BA 0.55. No architecture available to us resolves the bounded contrast into higher classification performance, and no amount of fine-tuning on pretrained features either discovers additional signal or reliably harms it — different architectures sit at different points of the 0.43–0.58 band, and the within-band ordering is not driven by representation capacity.

§6 has already identified this contrast as the operative variable by the controlled EEGMAT vs Stress paired comparison; §7 is the complementary ruling-out: once the contrast is bounded, architectural scale and pretraining source do not unbound it.

7.3 A constructive counter-check on EEGMAT

The §7 finding — architecture-independence of the ceiling — is a specifically *ceiling-regime*

statement. It does not claim that architectural choice never matters; it claims that architectural choice does not rescue a contrast-bounded task. On EEGMAT, where the within-subject contrast is neurally anchored, the same claim emphatically does not hold: the three frozen FM LPs reach 0.72–0.74 (§6.2), fine-tuning adds no more (Table 6.2), and the contrast-driven separability is at a qualitatively different operating point from the Stress ceiling.

Combining §6 (contrast-strength governs within-subject discrimination in frozen representations) with §7 (architecture does not bypass the ceiling once it is in force) gives a two-regime picture of EEG-FM behaviour on clinical rsEEG:

- **Contrast-anchored regime** (EEGMAT rest/arithmetic, ADFTD alpha-slowing): frozen representations already separate state; FT adds little; architecture and pretraining do contribute information but gains are modest on top of LP.
- **Contrast-absent or contrast-bounded regime** (per-recording DASS on UCSD Stress, MDD on TDBRAIN under CBraMod/REVE): no architecture within the range we can evaluate exceeds the cohort's intrinsic ceiling, and fine-tuning can degrade below frozen LP.

This two-regime behaviour is the operational content of the subject-dominance limit.

7.4 Implications for benchmark design

One practical consequence of §7 is that **benchmark improvements reported on contrast-bounded datasets are likely to be dominated by non-architectural factors** (evaluation protocol, label coding, HP search budget, seed count) rather than by architectural advances. §5 already showed that a single protocol correction on UCSD Stress absorbs a 20+ pp reported gap. §7 shows the converse: if a newly proposed architecture claims a meaningful improvement on a cohort that is structurally ceiling-bounded, the SDL diagnostic predicts that the improvement should either (a) be smaller than the single-model seed std once multi-seed replication is performed, or (b) appear specifically because the new architecture has access to a contrast anchor that the previous architectures did not.

The EEG Foundation Challenge 2025 [Truong et al., arXiv:2506.19141] frames cross-subject clinical psychopathology prediction as the open problem for this field. §7 suggests that pre-benchmark contrast anchoring (§9 Check 1) is a scientifically more productive move than further architectural iteration within the subject-dominance ceiling.

§8 asks the remaining within-ceiling question: granted that FMs do not escape the Stress ceiling on a single cohort, what does the picture look like across a broader 3-FM × 4-dataset grid under the same matched protocol?

SDL Paper Draft — §8 Cross-Dataset FT vs Frozen LP Analysis (v2, 2026-04-20)

Role in SDL thesis: *the generalisation audit.* §5 established the UCSD Stress ceiling; §6 showed EEGMAT state signal is linearly separable from frozen FM embeddings while Stress's is not; §7 showed architecture scale does not bypass the Stress ceiling. §8 asks whether the FT-vs-LP pattern observed on Stress generalises to three other clinical rsEEG cohorts, and whether the cases where FT helps admit a mechanistic explanation.

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§8 Fine-Tuning Rarely Exceeds Frozen Linear Probing Across Datasets

8.1 Grid: 3 FMs × 4 datasets, consistent per-FM protocol

We compare each FM's frozen linear probe (LP, per-window training + prediction pooling, §5.2 matched protocol, 8 seeds) against its own fine-tuning (FT, per-FM canonical recipe, 3 seeds) on the same four cohorts used throughout the paper: UCSD Stress (17 subjects, 70 recordings, DASS per-recording binary), ADFTD (65 subjects, 195 recordings, AD/FTD vs HC), TDBRAIN (359 subjects, 734 recordings, MDD vs HC), and EEGMAT (36 subjects, 72 recordings, rest vs mental arithmetic).

All twelve (FM × dataset) cells use **per-FM canonical recipes** (LaBraM `lr=1e-5, encoder_lr_scale=0.1, zscore`; CBraMod `lr=1e-5, elrs=0.1, none`; REVE `lr=3e-5, elrs=0.1, none`) — **without** per-dataset hyperparameter tuning. This reflects a realistic deployment scenario where clinical researchers cannot afford per-task HP search, and it structurally avoids test-set HP contamination. Source tables: `paper/figures/source_tables/master_frozen_ft_table_v2.{md,json}`.

8.2 Twelve-cell result

Table 8.1. Frozen LP (per-window, 8 seeds) vs Fine-tune (canonical recipe, 3 seeds). Balanced accuracy mean ± sample std (ddof = 1); Δ in percentage points.

Model	Phase	Stress	ADFTD	TDBRAIN	EEGMAT
LaBraM	Frozen LP	0.525 ± 0.040	0.653 ± 0.018	0.752 ± 0.011	0.736 ± 0.025
	Fine-tune	0.443 ± 0.083	0.709 ± 0.014	0.665 ± 0.045	0.731 ± 0.021
	ΔFT − LP	−8.1	**+5.6**	−8.7	−0.5
CBraMod	Frozen LP	0.430 ± 0.033	0.570 ± 0.030	0.525 ± 0.022	0.722 ± 0.022
	Fine-tune	0.548 ± 0.031	0.537 ± 0.027	0.488 ± 0.014	0.620 ± 0.058
	ΔFT − LP	**+11.8**	−3.3	−3.7	−10.2
REVE	Frozen LP	0.441 ± 0.022	0.652 ± 0.027	0.519 ± 0.022	0.733 ± 0.021
	Fine-tune	0.577 ± 0.051	0.658 ± 0.030	0.488 ± 0.019	0.727 ± 0.035
	ΔFT − LP	**+13.7**	+0.6	−3.1	−0.6

Summary: 3 of 12 cells show FT > LP by more than 1 pp, 3 are tied within 1 pp, and 6

show FT < LP by more than 1 pp. Median $\Delta = -3.1$ pp, mean $\Delta = -1.8$ pp. Fine-tuning is, on average, not a net-positive procedure across these datasets at these cohort sizes.

This pattern — fine-tuning failing to exceed a frozen linear probe under honest subject-disjoint evaluation — has been previously reported on individual datasets by BENDR (Kostas et al. 2021, 4 of 5 LOSO datasets) and AdaBrain-Bench (Wang et al. 2025, EEGPT on BCI-IV-2a cross-subject). Table 8.1 replicates the pattern across three FMs and four rsEEG cohorts under the matched per-window protocol. It is not a new phenomenon; it is an underreported one that headline tables in LaBraM/CBraMod/REVE papers, which use trial-level or fixed subject-disjoint single-split protocols, systematically understate.

8.3 Three rescue cases — two mechanisms

Only three of twelve cells show fine-tuning exceed frozen LP, and they split cleanly into two mechanistic categories.

Category A — Contrast-driven rescue (1 of 3): *LaBraM × ADFTD (+5.6 pp).* ADFTD is the dataset with the strongest published within-recording neural signature in our set — posterior alpha slowing in Alzheimer's disease is well-characterised [Babiloni 2020, Ieracitano 2021] and directly detectable in resting-state PSD. Among the three FMs, only LaBraM — the only architecture with a vector-quantised tokenizer — succeeds at projecting this contrast onto its classification head during FT. CBraMod and REVE FT on ADFTD **degrade** relative to their own frozen LP (-3.3, $+0.6$ pp) despite operating on the same signal. The rescue is therefore jointly contingent on (a) dataset contrast strength and (b) architectural plasticity. This matches the variance decomposition evidence in §4.5: LaBraM × ADFTD showed the largest Δlabel_frac under FT ($+10.65$ pp, unique among our 12 cells).

Category B — Small-cohort between-subject "rescue" (2 of 3): *CBraMod × Stress (+11.8 pp)* and *REVE × Stress (+13.7 pp).* Both rescue cases land on the smallest cohort in our study — 17 subjects / 70 recordings / 14 positives — and both are on architectures whose FT does not help on any of the three larger cohorts. Within Stress, positive recordings are structurally concentrated in a subset of subjects rather than uniformly distributed across all 17, which introduces a design-level collinearity between subject identity and recording-level label.

We therefore must ask: are these rescues genuine state-signal acquisition, or are they subject-identity shortcut exploitation under label-subject collinearity? §5.3 already showed that on the same cohort LaBraM FT is statistically indistinguishable from label-shuffle null ($p = 0.70$ under permutation over 10 seeds). Extending this permutation-null test to CBraMod and REVE on Stress is the cleanest way to separate the two hypotheses; that extension is pending compute and will be reported in an updated draft (§10 open items).

The SDL-consistent interpretation is that CBraMod and REVE fine-tuning is exploiting the 14-positive-recording class imbalance combined with subject-label collinearity — giving a classifier head the shortcut path "predict based on which subject-fingerprint cluster the test embedding is closest to". The class-imbalanced subset allows this shortcut to produce a 12–14 pp BA gain that a subject-invariant state signal could not. If the pending permutation-null test confirms CBraMod FT > CBraMod LP under shuffled Stress labels, the "rescue" is mechanically identified as shortcut exploitation. If not, the finding is unexplained and worth reporting as-is.

8.4 What this says about each FM

Across the 4-dataset grid, each FM has its own FT pattern:

- **LaBraM**: 1 genuine contrast-driven rescue (ADFTD), 2 tied-or-degraded (EEGMAT, TDBRAIN), 1 degraded (Stress). LaBraM's vector-quantised tokenizer and z-score preprocessing appear to give it the plasticity to succeed on strong-contrast cohorts, at the cost of FT stability on weaker-contrast or small cohorts.
- **CBraMod**: 0 contrast-driven rescues, 1 shortcut-suspect rescue on Stress, 3 degradations. On all three strong-contrast cohorts (ADFTD, EEGMAT, TDBRAIN), CBraMod FT degrades frozen LP. This is the most BENDR-style "FT is harmful" profile in our set.
- **REVE**: 0 contrast-driven rescues, 1 shortcut-suspect rescue on Stress, 2 ties, 1 degradation. The largest FM (∼1.4 B parameters) has the narrowest FT gain range across these datasets, consistent with scaling not helping and possibly amplifying FT instability (as AdaBrain-Bench 2025 reports on EEGPT for similar cross-subject shift).

None of these three patterns supports a general "fine-tuning strengthens EEG FMs" story at the 17–359-subject cohort scale.

8.5 Robustness checks

Two robustness concerns about the §8.2 table were pre-registered:

1. *Could the ceiling reflect our CV protocol?* We re-ran the LP arm on Stress with leave-one-subject-out (LOSO, 17 folds) — a tighter partitioning than 5-fold for small-N. LOSO shifts frozen LP BA by -1.2 to -3.3 pp across the three FMs, does not flip any verdict, and preserves the LaBraM > REVE > CBraMod ranking on Stress. Numbers: LaBraM 0.491, CBraMod 0.402, REVE 0.429 (LOSO, deterministic). `results/studies/perwindow_lp_all/stress/{model}_loso.json`.

2. *Could the LaBraM × Stress FT underperformance reflect a poor HP choice?* The full per-FM Stress HP sweep (`results/hp_sweep/20260410_dass/`, 6 HP combinations × 3 seeds) places the canonical recipe (lr = 1e-5) 5th of 6 for LaBraM. The best-performing HP (lr = 1e-4, encoder_lr_scale = 1.0) raises LaBraM Stress FT from 0.443 to 0.524 — still within 1 pp of LaBraM frozen LP (0.525) on the same cohort. Even under best-HP selection, LaBraM FT does not exceed frozen LP on Stress. (CBraMod and REVE best-HP = canonical recipe, so their numbers are unchanged.) The ceiling is not an HP artifact. See supplementary Table S2 for the full 54-run grid.

8.6 Take-away for clinical practice

For clinical researchers considering fine-tuning an EEG FM on their own small rsEEG cohort, this section's evidence says: **frozen linear probing is the safe default, and fine-tuning is a gamble unless the target label has a published within-recording neural anchor**. The single robust rescue we observed (LaBraM × ADFTD) coincides with the only cohort in our grid whose target contrast is neurally well-characterised. Fine-tuning on trait-like or state-weak labels (DASS, MDD) shows systematically negative or null returns, and apparent rescues on small imbalanced cohorts (Stress CBraMod/REVE) require mechanistic verification before being trusted.

§9 operationalises this guidance into a three-step pre-benchmark checklist for clinical EEG practitioners.

Appendix 8.A — Full master table (FT vs frozen LP) source reference

Master table v2: Frozen LP (per-window) vs FT (canonical per-FM recipe)

Date: 2026-04-20. Replaces `master_frozen_ft_table.md` (v1).

Protocol

- **Frozen LP**: per-window LogReg matched to FT's per-window training + prediction-pooling; percentile-clip + StandardScaler; `liblinear` solver; `StratifiedGroupKFold(5)` subject-disjoint; 8 seeds.
- **Fine-tuning**: per-FM canonical recipe — LaBraM `(lr=1e-5, encoder_lr_scale=0.1, zscore)`; CBraMod `(lr=1e-5, elrs=0.1, none)`; REVE `(lr=3e-5, elrs=0.1, none)`. Same recipe across all 4 datasets (no per-dataset HP tuning). 3 seeds.
- **Stress LaBraM FT source**: `exp03_stress_erosion/ft_real/` (canonical lr = 1e-5), **NOT** `hp_sweep/20260410_dass/` best-HP (lr = 1e-4). Consistency across datasets is the criterion.
- Balanced accuracy; sample std (ddof = 1).

Results table

Model	Phase	Stress	ADFTD	TDBRAIN	EEGMAT
LaBraM	Frozen LP	0.525 ± 0.040 (n = 8)	0.653 ± 0.018 (n = 8)	0.752 ± 0.011 (n = 8)	0.736 ± 0.025 (n = 8)
	Fine-tune	0.443 ± 0.083 (n = 3)	0.709 ± 0.014 (n = 3)	0.665 ± 0.045 (n = 3)	0.731 ± 0.021 (n = 3)
CBraMod	Frozen LP	0.430 ± 0.033 (n = 8)	0.570 ± 0.030 (n = 8)	0.525 ± 0.022 (n = 8)	0.722 ± 0.022 (n = 8)
	Fine-tune	0.548 ± 0.031 (n = 3)	0.537 ± 0.027 (n = 3)	0.488 ± 0.014 (n = 3)	0.620 ± 0.058 (n = 3)
REVE	Frozen LP	0.441 ± 0.022 (n = 8)	0.652 ± 0.027 (n = 8)	0.519 ± 0.022 (n = 8)	0.733 ± 0.021 (n = 8)
	Fine-tune	0.577 ± 0.051 (n = 3)	0.658 ± 0.030 (n = 3)	0.488 ± 0.019 (n = 3)	0.727 ± 0.035 (n = 3)

Δ (FT − Frozen LP), percentage points

Model	Stress	ADFTD	TDBRAIN	EEGMAT
LaBraM	-8.1	+5.6	-8.7	-0.5
CBraMod	+11.8	-3.3	-3.7	-10.2
REVE	+13.7	+0.6	-3.1	-0.6

Verdict count (12 cells)

- **FT > LP ($\Delta > +0.01$)**: 3 cells
- **FT ≈ LP ($|\Delta| \leq 0.01$)**: 3 cells
- **FT < LP ($\Delta < -0.01$)**: 6 cells

Per-cell verdicts

Cell	Δ	Verdict
LaBraM × Stress	-8.1 pp	↓ FT degrades
LaBraM × ADFTD	+5.6 pp	↑ FT rescues
LaBraM × TDBRAIN	-8.7 pp	↓ FT degrades
LaBraM × EEGMAT	-0.5 pp	≈ tied
CBraMod × Stress	+11.8 pp	↑ FT rescues
CBraMod × ADFTD	-3.3 pp	↓ FT degrades
CBraMod × TDBRAIN	-3.7 pp	↓ FT degrades
CBraMod × EEGMAT	-10.2 pp	↓ FT degrades
REVE × Stress	+13.7 pp	↑ FT rescues
REVE × ADFTD	+0.6 pp	≈ tied
REVE × TDBRAIN	-3.1 pp	↓ FT degrades
REVE × EEGMAT	-0.6 pp	≈ tied

Key differences vs v1

Cell	v1 LP (feature-avg 8s)	v2 LP (per-window 8s)	v1 FT	v2 FT	Notes
LaBraM × Stress	0.605 ± 0.032	**0.525 ± 0.040**	0.524 ± 0.010	**0.443 ± 0.083**	v1 LP was feat-avg (inflated), FT was hp_sweep best-HP (lr = 1e-4) not canonical
CBraMod × Stress	0.452 ± 0.032	**0.430 ± 0.033**	0.548 ± 0.031	0.548 ± 0.031	LP per-window lower; FT identical
REVE × Stress	0.494 ± 0.018	**0.441 ± 0.022**	0.577 ± 0.051	0.577 ± 0.051	LP per-window lower; FT identical
Others (9 cells)	old 3-seed feat-avg	**new 8-seed per-window**	unchanged	unchanged	LP more rigorous, FT unchanged

Dataset metadata

Dataset	Task	N subjects	N recordings	Labels
Stress	DASS stress state	17	70	14 pos / 56 neg (per-recording)

ADFTD	AD vs HC	65	195	108 pos / 87 neg
TDBRAIN	MDD vs HC	359	734	640 pos / 94 neg
EEGMAT	rest vs arithmetic task	36	72	36 rest / 36 task (paired within-subject)

SDL Paper Draft — §9 A Pre-Flight Checklist for Clinical EEG-FM Deployment (v2, 2026-04-20)

Role in SDL thesis: *the operational output.* The preceding sections establish (a) that evaluation-protocol corrections can collapse headline EEG-FM benchmarks by 20–40 pp (§5); (b) that frozen LP separability, not fine-tuning gain, is the primary diagnostic of whether an EEG FM will be useful on a new cohort (§6); (c) that aperiodic 1/f structure causally carries subject identity in FM embeddings (§6.5); (d) that cross-dataset fine-tuning shows 1 genuine rescue, 3 ties, and 6 degradations across 12 cells (§8). §9 translates these findings into a concrete, actionable decision rule for clinical researchers contemplating EEG-FM use on their own cohort.

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§9 A Pre-Flight Checklist for Clinical EEG-FM Deployment

9.1 Who this section is for

Three audiences:

1. **Clinical researchers** with a small-to-medium rsEEG cohort (20–500 subjects) who want to know whether an EEG foundation model will help on their data.
2. **Methodologists** reviewing EEG-FM papers and looking for concrete red-flags that indicate reported accuracies will not generalise.
3. **EEG-FM developers** choosing benchmarks against which to validate architectural choices.

All three questions reduce to the same core: under honest subject-disjoint evaluation, will this (FM × cohort × label) combination yield a clinically useful model, and if not, why not?

9.2 The checklist (three decisions)

Before investing compute in fine-tuning an EEG FM on a clinical rsEEG cohort, work through the three checks below in order. Each gate is designed to be cheap to run, producing a binary or ordinal signal that determines whether the next gate is worth passing through.

Check 1 — Does the label have a published within-recording neural anchor?

Purpose: Separates labels that have a known-detectable spectral or temporal signature from labels that are trait-level aggregations without direct per-recording neural correlate.

How: Search the neurophysiology literature for the task or diagnostic category. A qualifying anchor is a peer-reviewed effect at the epoch/recording level — e.g., alpha desynchronisation for mental arithmetic [Klimesch 1999], posterior alpha slowing for Alzheimer's [Babiloni 2020], beta-power reduction for Parkinson's [Cassani 2018]. A non-qualifying anchor is a group-level longitudinal correlation with no published per-recording EEG signature — e.g., DASS-21 score, Big Five personality factors, depression severity on long timescales.

Expected outcome on our grid:

- ADFTD (alpha slowing): ✓ qualifying anchor
- EEGMAT (rest vs arithmetic, alpha ERD): ✓ qualifying anchor
- TDBRAIN MDD (heterogeneous; weak within-recording anchor): ✗ non-qualifying
- Stress DASS per-recording: ✗ non-qualifying (no published neural correlate at the relevant temporal resolution)

Decision: If Check 1 fails, be prepared to accept classification ceilings of 0.50–0.58 BA under honest evaluation (our Stress finding). Don't invest in fine-tuning. Classical band-power baselines are likely within 5 pp of anything an FM will achieve on this label. Consider collecting additional structured tasks or choosing a label with a published anchor.

Check 2 — What is the frozen linear probe balanced accuracy under subject-disjoint CV?

Purpose: Measures whether the FM's frozen representation already contains linearly separable state signal for this label.

How: Extract frozen features per-window from the FM (using its per-FM input norm — this matters; see CLAUDE.md §2). Train logistic regression on per-window training-fold samples with class balancing, pool test-fold window predictions per recording, compute recording-level BA under `StratifiedGroupKFold(5)` with `groups=patient_id`. Repeat over at least 5 seeds. This takes < 1 hour on CPU once features are extracted; feature extraction takes < 1 hour on a single consumer GPU for 100–500-subject cohorts.

Expected outcome on our grid:

- EEGMAT: LP BA $= 0.72$–0.74 across three FMs (strong signal present)
- ADFTD: LP BA $= 0.57$–0.65 (moderate)
- TDBRAIN: LP BA $= 0.52$–0.75 (varies wildly by FM: LaBraM strong, CBraMod/REVE near chance)
- Stress: LP BA $= 0.43$–0.53 (below classical band-power baseline 0.553)

Decision: LP BA > 0.70 → the frozen representation already works; deploy frozen LP. LP BA in $[0.55, 0.70]$ → there is some signal; fine-tuning may rescue (see Check 3). LP BA < 0.55 → no linearly separable state signal; do not fine-tune (you will either overfit or exploit subject shortcut).

Check 3 — What is the subject-ID decodability of the frozen embedding, and how does it compare to state-label decodability?

Purpose: Flags cases where apparent signal is actually subject-identity memorisation rather than state discrimination. This is the pre-flight diagnostic for the subject-dominance limit.

How: Train a multi-class logistic regression on per-window features to predict `patient_id`, using session-held-out split within each subject (half of each subject's recordings for training, other half for test). Pool per-window probabilities to recording level. Compute multi-class balanced accuracy; the chance level is 1/N_subjects. Compare to the state-label LP BA from Check 2.

Interpretation:

- If subject-ID BA is $> 10\times$ chance (e.g., 0.50 for $N = 20$ subjects, where chance $= 0.05$), the frozen FM embedding is a strong subject fingerprint. This is structural — §4

shows this is true for LaBraM/CBraMod/REVE on every dataset we tested. The informative number is the *ratio* of state-label BA to subject-ID decodability.
- State-label BA $>>$ subject-ID BA (chance-adjusted) → state signal is well-separated from subject identity. FT has scope to add value.
- State-label BA $\approx$ subject-ID BA → state and subject are entangled. FT on this label risks exploiting subject shortcut rather than state signal.

Supplementary diagnostic (FOOOF ablation, §6.5): For research-grade evaluation, verify that subject-ID BA drops substantially when aperiodic 1/f is removed from the input signal. A drop $>$ 10 pp confirms the standard pattern (subject identity encoded in aperiodic component, periodic oscillations carry state). Absence of this drop indicates an atypical FM representation and warrants caution.

Decision: If state-label and subject-ID BAs are within 1.5$\times$ of each other, treat fine-tuning gains on between-subject labels as potentially spurious. Run a permutation-null control: fine-tune with shuffled labels. If shuffled-label BA is not significantly lower than real-label BA, your rescue is shortcut, not signal.

9.3 Applying the checklist to our datasets

Dataset	Check 1 (Anchor)	Check 2 (Frozen LP BA)	Check 3 (Subj-ID vs state-label)	Verdict
EEGMAT (rest/task)	✓ Alpha ERD	0.72–0.74	State BA $>$ Subj-ID BA	**Deploy frozen LP; FT does not add value but does not harm**
ADFTD (AD vs HC)	✓ Alpha slowing	0.57–0.65	Comparable	**FT on LaBraM only (vector-quantised tokenizer has plasticity here); frozen LP on others**
TDBRAIN (MDD vs HC)	✗ No published anchor	0.52–0.75 (LaBraM-only)	Not checked	**Frozen LP on LaBraM; do not fine-tune**
Stress (DASS per-rec)	✗ No published anchor	0.43–0.53	**Subject-ID $\gg$ state-label**	**Do not fine-tune; do not expect clinical utility at this cohort size with this label**

9.4 What this changes for the field

The checklist operationalises two findings that the EEG-FM benchmark literature has not yet internalised:

- **Frozen LP is a diagnostic, not a weakness.** Most EEG-FM papers report LP as a secondary baseline to fine-tuning, sometimes downplayed when FT wins. Under honest evaluation on small clinical cohorts, LP is often the more informative number: it reveals whether the FM's pretraining has given the downstream task any linear separability at all. If LP is high, LP is enough; if LP is low, fine-tuning is gambling.

- **Subject-ID probe is the pre-flight diagnostic for shortcut risk.** Check 3 adds a single sklearn pipeline to any EEG-FM evaluation that turns "our FT BA went up 12 pp" from a celebration into a diagnostic signal: did the 12 pp come from state separation or from subject-fingerprint shortcut? With N $\leq$ 100 subjects this question is non-trivial and

easy to get wrong.

We encourage future EEG-FM papers reporting cross-subject clinical benchmarks to report Check 2 and Check 3 explicitly, alongside the primary fine-tuning numbers.

9.5 Limitations of the checklist

The checklist is a deployment-time diagnostic, not a research-time ceiling. It assumes:

- The cohort uses a standard rsEEG montage ($\geq$ 19 channels, standard 10–20).
- Labels are binary or low-cardinality categorical.
- The research question is cross-subject prediction, not within-subject tracking.

For within-subject longitudinal tracking or within-session state detection tasks (e.g., drowsiness over an 8-hour shift), different considerations apply; the within-subject contrast literature [Klimesch 1999; Pfurtscheller 1997] is more directly applicable than any EEG-FM. Where subject identity can be controlled by design (same subject, multiple conditions), SDL does not bind.

For very large cohorts (N > 10ˆ4 subjects, which exist in some industrial contexts), fine-tuning dynamics may differ, and the pattern reported here may not transfer.

SDL Paper Draft — §10 Limitations, Open Questions, and Future Work (v2, 2026-04-20)

Role in SDL thesis: *the honest accounting.* Every critique paper must itself be audit-able. §10 enumerates the claims we do not fully support, the experiments we deferred, and the dimensions along which the pre-flight checklist of §9 may need revision.

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§10 Limitations, Open Questions, and Future Work

10.1 Wang et al. 2025 reproduction is incomplete

Our §5.1 audit replicates the protocol-induced inflation on UCSD Stress (trial-level → subject-level CV accounts for approximately 20 pp of the 40 pp gap between Wang et al. 2025's reported 0.9047 BA and our honest-evaluation 0.443–0.577). The residual ∼20 pp we attribute to unspecified design choices in the Wang pipeline (hyperparameter selection, label coding threshold, window policy, possibly features selection) that we cannot audit without full training logs. We do not claim full reproduction of the Wang pipeline. Our claim is narrower: under our controlled matched protocol, the within-pipeline trial-vs-subject inflation alone does not account for 0.9047-level accuracy, and no architecture we tested reaches 0.9047 under honest CV.

10.2 Stress CBraMod and REVE FT rescue — permutation-null control pending

Two of the three "FT > LP" cells in our §8 twelve-cell grid are on the small Stress cohort (CBraMod × Stress +11.8 pp, REVE × Stress +13.7 pp). Stress has 14 positive recordings distributed non-uniformly across 17 subjects, which creates a label-subject collinearity that could allow fine-tuning to exploit subject-identity shortcut rather than state-discriminative signal.

We have run permutation-null control for LaBraM × Stress (§5.3, $p = 0.70$ one-sided) — real-label FT is statistically indistinguishable from shuffled-label FT. We have *not* yet run the equivalent control for CBraMod × Stress and REVE × Stress. Our current reporting therefore treats these rescues as *mechanistically suspect* rather than confirmed shortcut. If a permutation-null test confirms that CBraMod/REVE FT on shuffled Stress labels still exceeds their own frozen LP, the "rescue" is mechanically identified as shortcut exploitation of label-subject collinearity; if not, the finding is genuine small-cohort FT-rescue and requires deeper investigation.

This test requires 10 permutation seeds × 2 FMs = 20 additional fine-tuning runs on Stress. Each run is compute-comparable to a single FT seed (∼15 minutes on a single GPU). Total: ∼5 GPU-hours. We defer this to the next revision.

10.3 Per-dataset HP sweep on ADFTD, TDBRAIN, EEGMAT not performed

We report FT under each FM's canonical per-paper recipe across all 4 datasets, *without* per-

dataset HP tuning. This choice is deliberate (§3.6 rationale) but has a known cost: FM-×-dataset cells where FT degrades LP (6 of 12 cells) could conceivably show different results under a per-dataset HP sweep with nested CV.

On UCSD Stress, where we do have a 6-HP × 3-seed sweep, the best-HP LaBraM FT (0.524) is statistically indistinguishable from the frozen LP (0.525) — so at least on this cohort, per-dataset HP tuning does not overturn the main claim. Extending the HP sweep to ADFTD, TDBRAIN, and EEGMAT (54 runs per dataset × 3 datasets = 162 additional FT runs) would strengthen the claim, at ~1–2 GPU-days. We defer this.

Importantly, reviewers skeptical of the FT < LP claim on ADFTD/TDBRAIN/EEGMAT should note that the canonical per-FM recipe used here is the same recipe the original LaBraM, CBraMod, and REVE papers report for their own downstream evaluations. If our recipe is "not best", it is at least the recipe those benchmark tables were produced with.

10.4 Architecture panel is Stress-only and does not cover all 2017–2025 models

The §7 architecture-independence result is on UCSD Stress only. Extending it to ADFTD / TDBRAIN / EEGMAT would require multi-seed training of EEGNet, ShallowConvNet, and similar models on those cohorts, which is straightforward but was not part of the present scope. The currently-missing architectures from §7's panel — DeepConvNet and EEGConformer — were evaluated in an earlier archived run (results/archive/2026-04-09_subject_dass_cleanup/) under the deprecated `--label subject-dass` protocol and require rerunning under the current `--label dass` protocol (< 1 hour each).

10.5 FOOOF ablation is on Stress + EEGMAT only

§6.5 reports the FOOOF aperiodic/periodic ablation on Stress (17 subjects) and EEGMAT (36 subjects). Extending this to ADFTD (65 subjects) and TDBRAIN (359 subjects) would strengthen the causal claim and is methodologically identical (script `scripts/analysis/fooof_ablation.py` already handles generic datasets). The principal scientific question is whether LaBraM's reduced aperiodic-dependence (§6.5 Table 6.3: LaBraM EEGMAT aperiodic-removed $\Delta = -2.8$ pp vs REVE's -26.0 pp) replicates on larger cohorts with different physiological signatures. Estimated compute: ~6 GPU-hours for the full ADFTD + TDBRAIN ablation + extraction + probe pipeline. Deferred to future work.

10.6 Small-cohort statistical reliability

UCSD Stress (70 recordings, 14 positive, 17 subjects) is the smallest cohort in our grid, and our §5 and §7 claims rest in part on its results. Multi-seed and multi-protocol (5-fold CV, LOSO) replication gives consistent BA estimates in the 0.43–0.58 band across all conditions we tested, and permutation-null control (on LaBraM) is not rejected at $p = 0.05$. However, the intrinsic variance of any DASS-related claim at $N = 70$ recordings is large, and no single study at this cohort size — including ours — produces confidence intervals tight enough to settle per-pp questions. The §5 claims should therefore be interpreted as "the reported 0.9047 is not reachable under honest evaluation", not as "the true ceiling is exactly 0.55".

Following the multisite reliability study of $N = 2{,}874$ rsEEG recordings [bioRxiv 2025.11.10.687610, 2025], results at $N < 300$ should be considered provisional.

10.7 Protocol assumptions

Our evaluations assume:

- **Subject-disjoint CV** as the honest evaluation protocol. This is standard in clinical EEG DL [Brookshire 2024] but is sometimes relaxed in emotion and cognitive-load benchmarks where within-subject performance is explicitly the target.
- **Recording-level labels** as the prediction target. Some clinical settings require window-level or sub-recording-level prediction (e.g., sleep staging). Our protocol does not directly address this regime.
- **A standard rsEEG montage** with $\geq$ 19 channels. Exotic montages or very low-density setups may behave differently.

10.8 LaBraM's reduced aperiodic-dependence — architectural explanation is speculative

§6.5 shows LaBraM's subject-ID probe drops only 2.8 pp under aperiodic removal, versus 14.2 pp (CBraMod) and 26.0 pp (REVE) on EEGMAT. We speculate in §6.5 that LaBraM's vector-quantised tokenizer combined with z-scored input normalisation may normalise aperiodic slope variation out of the tokens, leaving the discrete token distribution less aperiodic-dependent. This is a plausible architectural explanation but we have not performed the controlled ablation that would separate tokenizer vs preprocessing contributions. An architectural ablation study (VQ vs no-VQ, zscore vs raw input) is a future-work direction.

10.9 Clinical deployment: our checklist is necessary, not sufficient

§9's three-step pre-flight checklist is designed as a minimum screening procedure before investing fine-tuning compute. Passing all three checks does not guarantee clinical utility; it only rules out a set of known failure modes (trait-like labels without neural anchor, LP-unseparable representations, subject-shortcut-dominated rescues). Achieving clinical utility further requires:

- Generalisation to unseen cohorts (external validation set).
- Sensitivity to cases where the model's prediction is disease-relevant (calibration, not just accuracy).
- Robustness to device, montage, and preprocessing shifts not captured by a single-corpus evaluation.

Our checklist is necessary for a clinical researcher to not waste compute on a broken benchmark; it is not sufficient to produce a clinical product.

10.10 Future work — three concrete directions

1. **Causal FOOOF intervention during pretraining.** The §6.5 ablation shows aperiodic 1/f is the substrate of subject dominance in pretrained FM embeddings. A natural next experiment is to **pretrain an FM on FOOOF-detrended EEG** (aperiodic removed at preprocessing time) and measure whether the resulting model has reduced subject dominance without sacrificing task-discriminative capacity. This is an architectural change with clean expected comparison against an un-detrended baseline of identical size and corpus.

2. **Subject-shortcut regularisation.** If subject dominance is partially driven by label-subject collinearity in small cohorts (the Stress CBraMod/REVE rescue hypothesis in §10.2), explicit subject-adversarial regularisation during fine-tuning on small-N clinical cohorts may recover the apparent signal without the shortcut. Methods already exist

[Özdenizci 2020, Zhang 2025]; they have not been integrated into EEG-FM fine-tuning pipelines.

3. **Within-subject longitudinal benchmarks.** Our pre-flight checklist targets cross-subject clinical prediction. A complementary research program would ask whether EEG FMs add value for within-subject longitudinal tracking (e.g., circadian drowsiness, pre/post-intervention state) where subject identity is fixed by design. The Stress cohort itself (longitudinal DSS per-recording trajectories) is a candidate; the 54-recording within-subject DSS-threshold analysis in our archive suggests within-subject FM performance is poor at $N \leq 10$ recordings per subject, but this has not been systematically benchmarked.